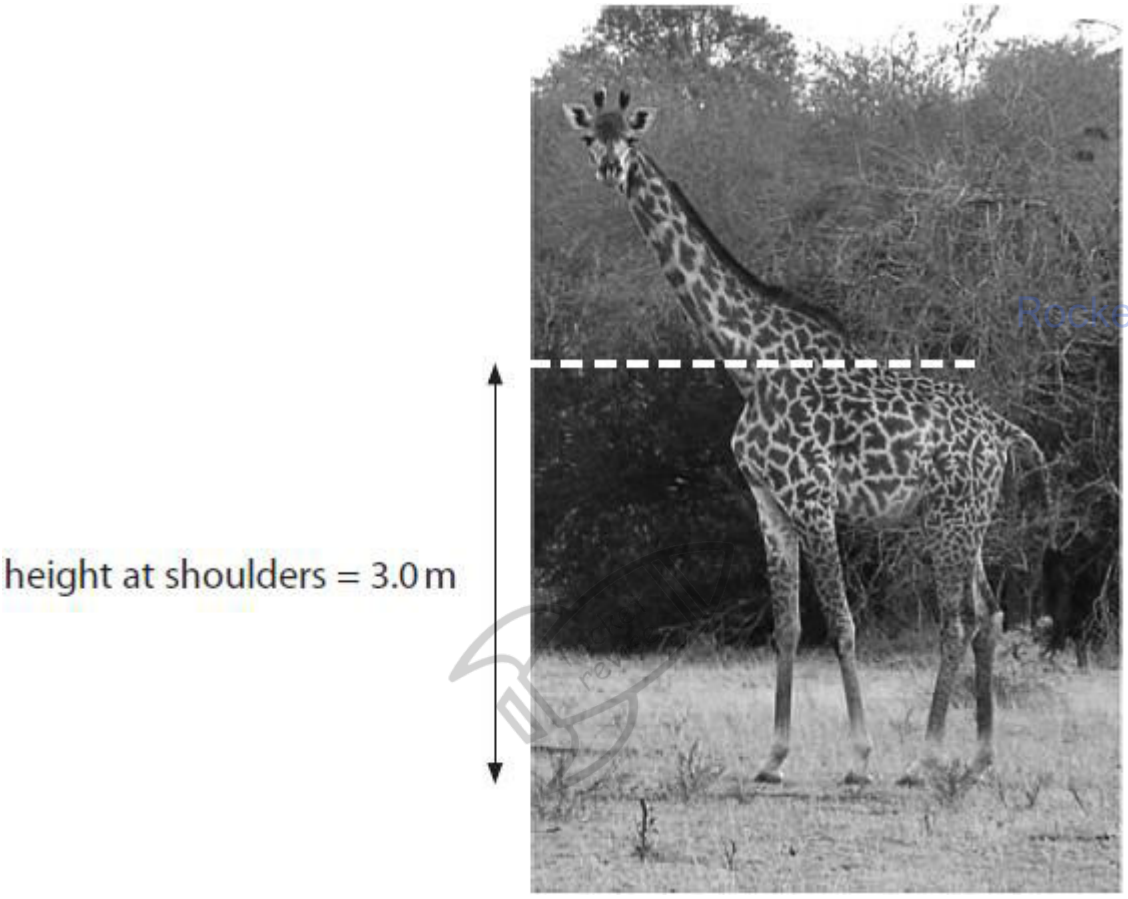
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Questions

Q1.

The photograph shows an adult giraffe.



(a) Estimate the length of the neck of this giraffe from the shoulders, using the information in the photograph.

(1)

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Answer ..... m

(b) The heart of an adult giraffe can be 60 cm long.

Explain why the heart of a giraffe needs to be so large.

(2)

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(c) The arteries near the heart of a giraffe are highly elastic.

(i) Draw a labelled diagram of an artery, as seen in section.

(3)

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(ii) Explain why the arteries near the heart of a giraffe are highly elastic.

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(d) Damage to the legs of the giraffe could result in excessive bleeding.

To prevent excessive bleeding, the capillaries near the surface of the skin are very narrow.

(i) Explain why very narrow capillaries prevent excessive bleeding.

(2)

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(ii) The red blood cells of the giraffe are about one third the size of human red blood cells, so that they can pass through the very narrow capillaries.
The small size of the red blood cells ensures that the legs of the giraffe have a good supply of oxygen.
Explain why smaller red blood cells increase the supply of oxygen to the legs.

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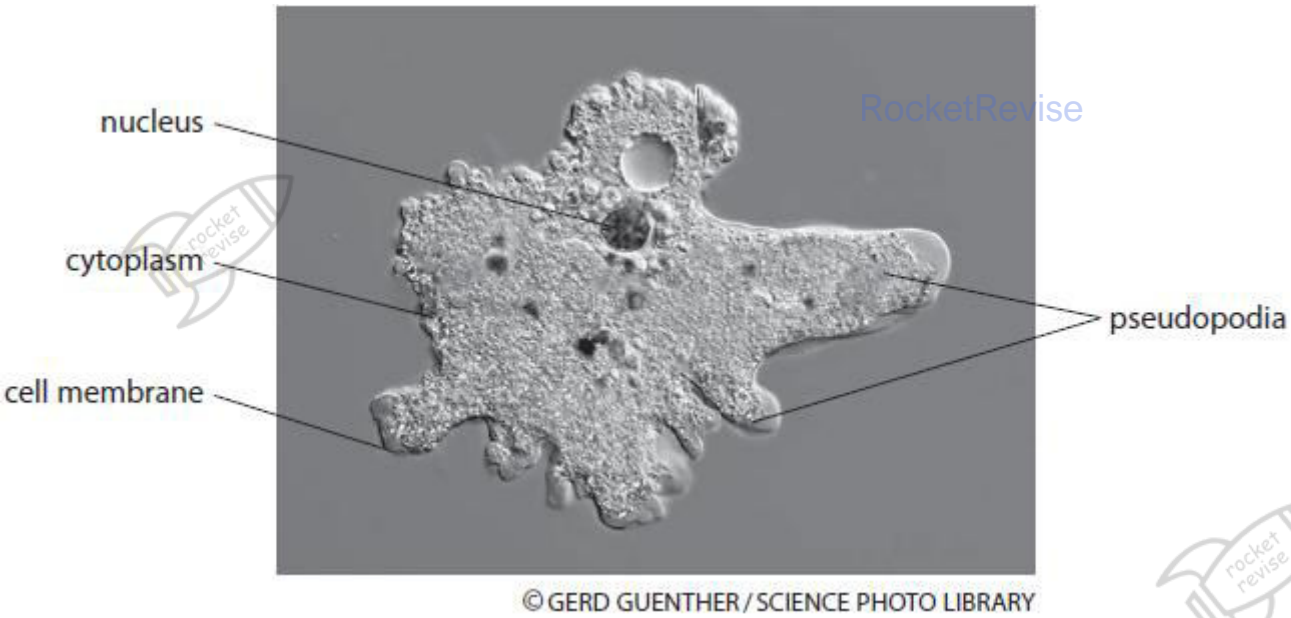
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(Total for question = 12 marks)

Q2.

An amoeba is a single-celled organism that lives in water.

The photograph shows an amoeba, as seen using a light microscope.



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Substances in the water can enter the amoeba by a variety of transport mechanisms.

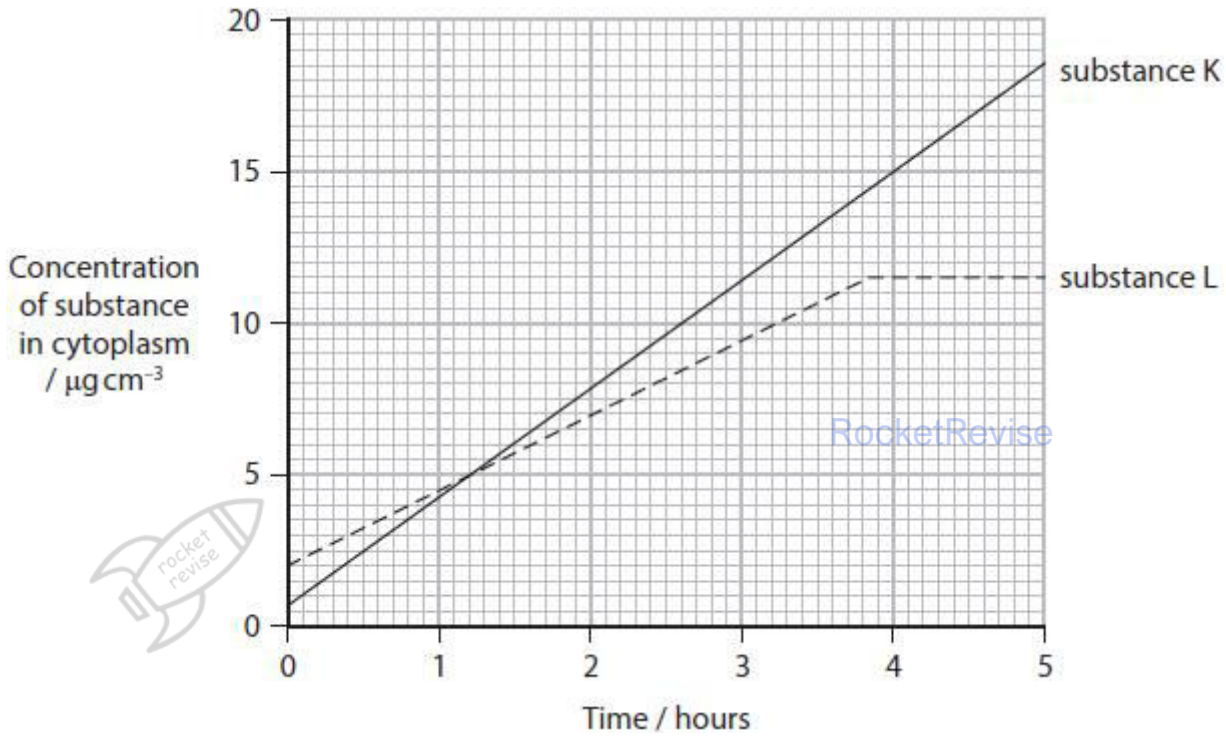
(a) The table shows some features of transport mechanisms.

Which feature is true for active transport only, facilitated diffusion only, both active transport and facilitated diffusion, or not true for both active transport and facilitated diffusion?

(3)

Feature	Transport mechanism			
	active transport only	facilitated diffusion only	both active transport and facilitated diffusion	not true for both active transport and facilitated diffusion
passive process	☐	☐	☐	☐
membrane proteins involved	☐	☐	☐	☐
direction of transport can be up the concentration gradient	☐	☐	☐	☐

(b) The graph shows the uptake of two substances, substance K and substance L, into the cytoplasm of an amoeba in water at a temperature of 18 °C.



(i) Explain the differences in the uptake of substance K and substance L.

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(ii) On the graph, draw a line to show the uptake of substance L at 10 °C.

(2)

(c) Pseudopodia are formed by cytoplasm flowing towards the membrane and changing its shape.

(i) Explain why the membrane is able to change its shape when cytoplasm flows towards it.

(2)

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(ii) Explain how the uptake of substances would be affected if the amoeba increased its number of pseudopodia.

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(Total for question = 11 marks)

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Q3.

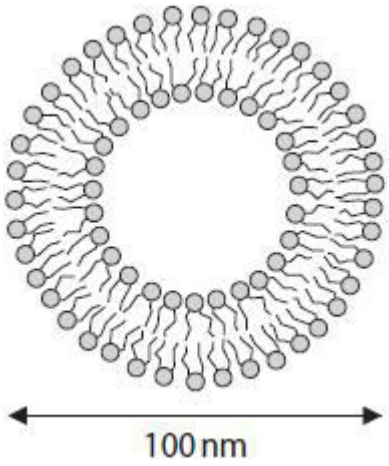
Liposomes are spherical structures composed of phospholipids. They can be made by adding phospholipids to water.

Liposomes can be used to study membrane permeability.

(a) The diagram shows a liposome.

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(i) Calculate the volume of this liposome, using the formula

$$V = \frac{4}{3}\pi r^3$$

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(2)

Answer .....

(ii) Explain the arrangement of phospholipids in liposomes.

(2)

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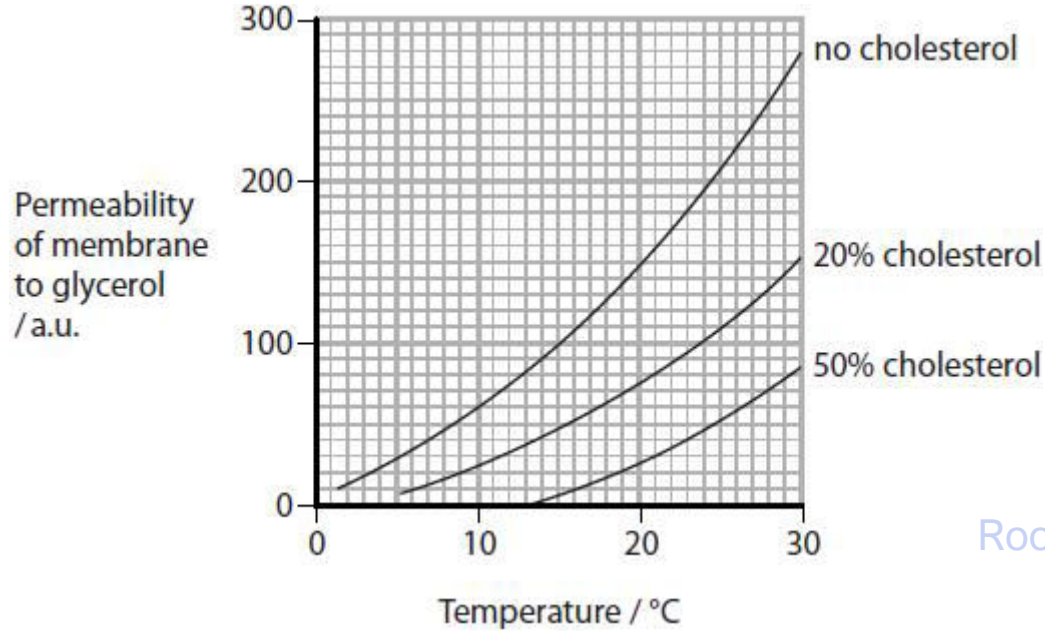
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(b) The presence of cholesterol in the membrane affects membrane permeability.

A student investigated the effect of cholesterol on the permeability of liposomes to glycerol at different temperatures.
Liposomes were made by replacing 20% and 50% of the phospholipid with cholesterol.
Liposomes without cholesterol were also made.
The graph shows the results of this investigation.

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(i) How does glycerol pass through the liposome membrane?

(1)

- ☒ **A** active transport
- ☒ **B** diffusion
- ☒ **C** endocytosis
- ☒ **D** osmosis

(ii) Describe the effects of cholesterol and temperature on membrane permeability, as shown in the graph.

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(iii) Explain why cholesterol and temperature affect membrane permeability.

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(Total for question = 10 marks)

Q4.

The photographs show two mammals, an elephant and a mouse.

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Magnification ×0.02

(a) The height of a mouse is 3 cm.

Calculate how many times taller an elephant is than a mouse.
Use the white line drawn on the photograph of the elephant to calculate this value.

(2)

Answer .....

(b) The respiratory system of an elephant is different from that of other mammals.

The lungs are attached to the chest cavity wall and diaphragm by collagen fibres.
Describe how the lungs of an elephant are adapted for gas exchange.

(3)

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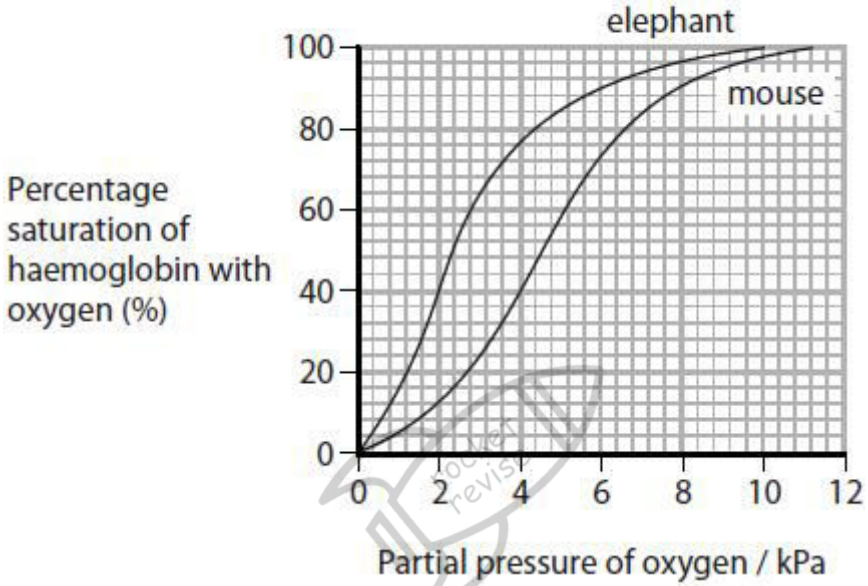
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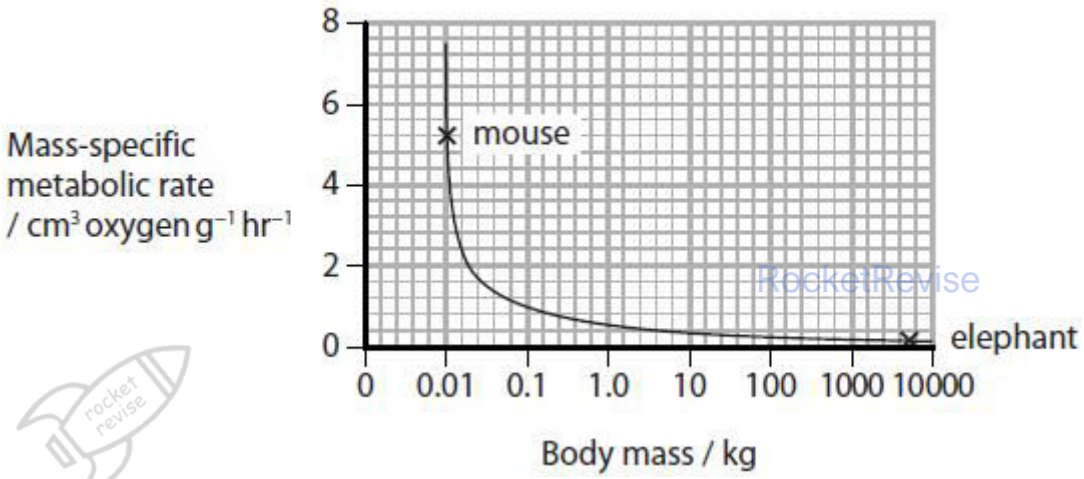
* (c) Graph 1 shows the oxygen dissociation curve of haemoglobin for a mouse and for an elephant.

Graph 1



Graph 2 shows the mass-specific metabolic rate for a mouse and for an elephant.
Mass-specific metabolic rate is a measure of how much oxygen is needed for chemical reactions per gram of body tissue.

Graph 2



Explain the difference in the oxygen dissociation curves of haemoglobin for a mouse and for an elephant.
Use the information in both graphs to support your answer.

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(Total for question = 11 marks)

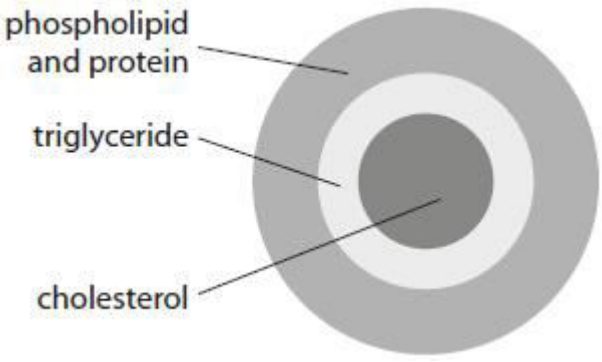
Q5.

Low-density lipoproteins (LDLs) transport lipids around the body in the blood.

Low-density lipoproteins can result in the development of atherosclerosis.

They can be absorbed into the endothelial cells lining arteries and broken down by free radicals.

The diagram shows a low-density lipoprotein containing cholesterol.



(a) Compare and contrast the structure of a triglyceride and a phospholipid.

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(b) Explain why the properties of LDLs enable cholesterol to be transported in the blood.

(3)

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(c) The diameters of LDLs range from 19 nm to 24 nm.

The table shows some information about LDLs.

Diameter of LDL / nm	Volume of LDL / nm^3	Volume of cholesterol / nm^3	Ratio of LDL volume to cholesterol volume
19	3590	523	7:1
24		523	

(i) Complete the table by calculating the volume of LDL and the ratio of LDL volume to cholesterol volume.

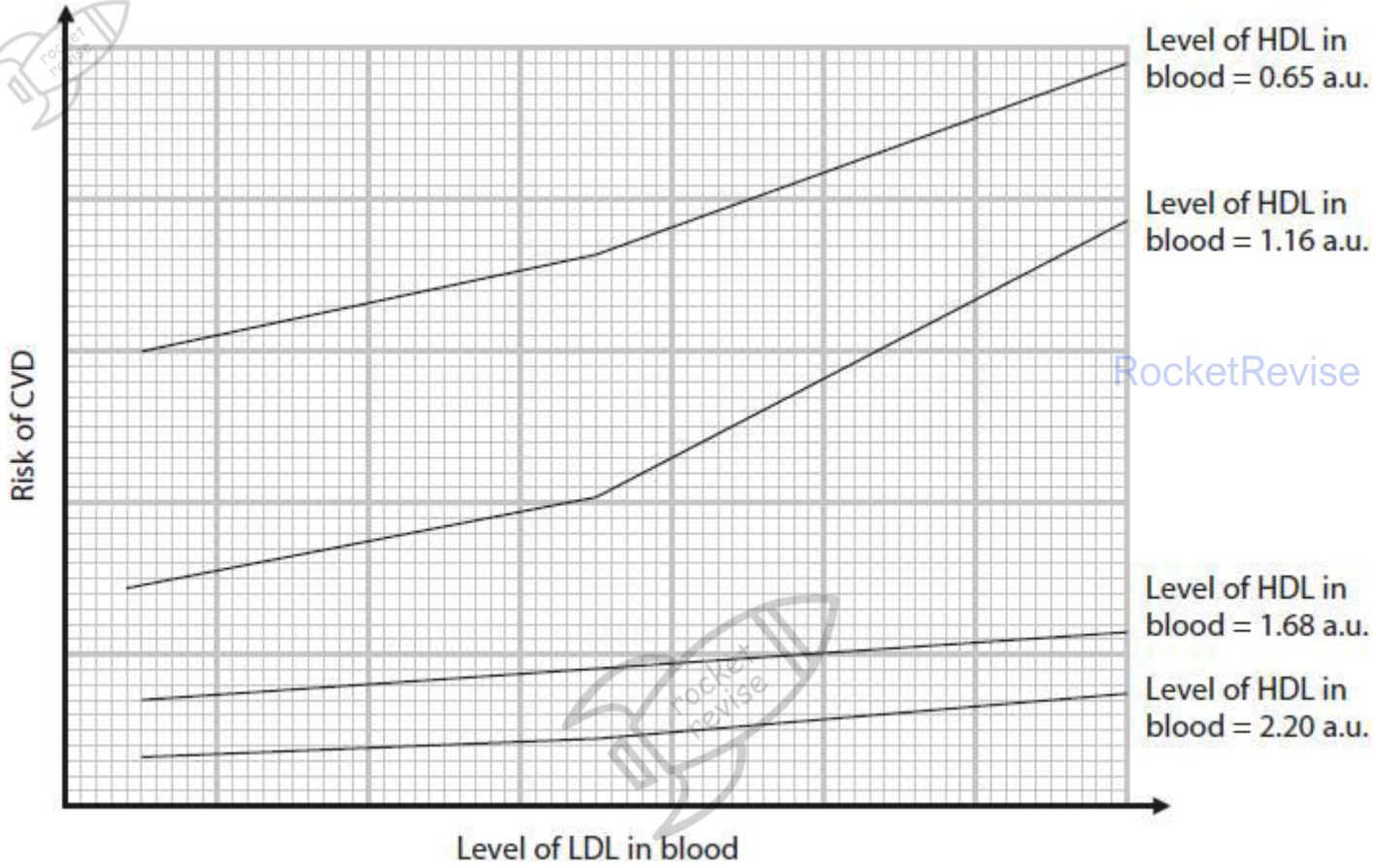
Use the formula $v = \frac{4}{3}\pi r^3$

(3)

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*(ii) The graph shows the relationship between LDLs, high-density lipoproteins (HDLs) and the risk of CVD.



Explain why measuring only the level of LDL in the blood is **not** a reliable predictor of CVD.
Use the graph, all the information in this question and your own knowledge to support your answer.

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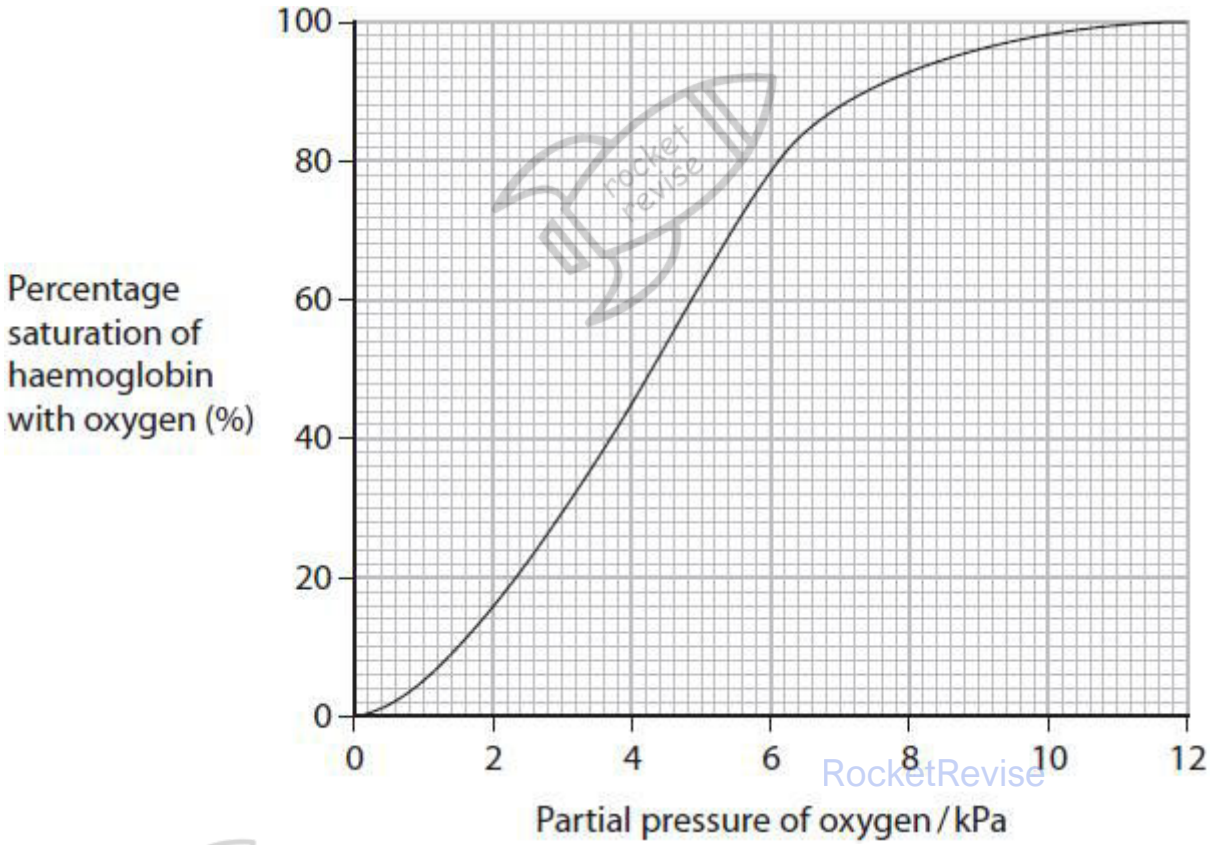
(Total for question = 15 marks)

Q6.

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One role of haemoglobin is to transport oxygen.

The graph shows an oxygen dissociation curve of haemoglobin.



(a) (i) State what is meant by the term partial pressure.

(1)

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(ii) Explain how the structure of haemoglobin causes the oxygen dissociation curve of haemoglobin to be this sigmoidal (S) shape.

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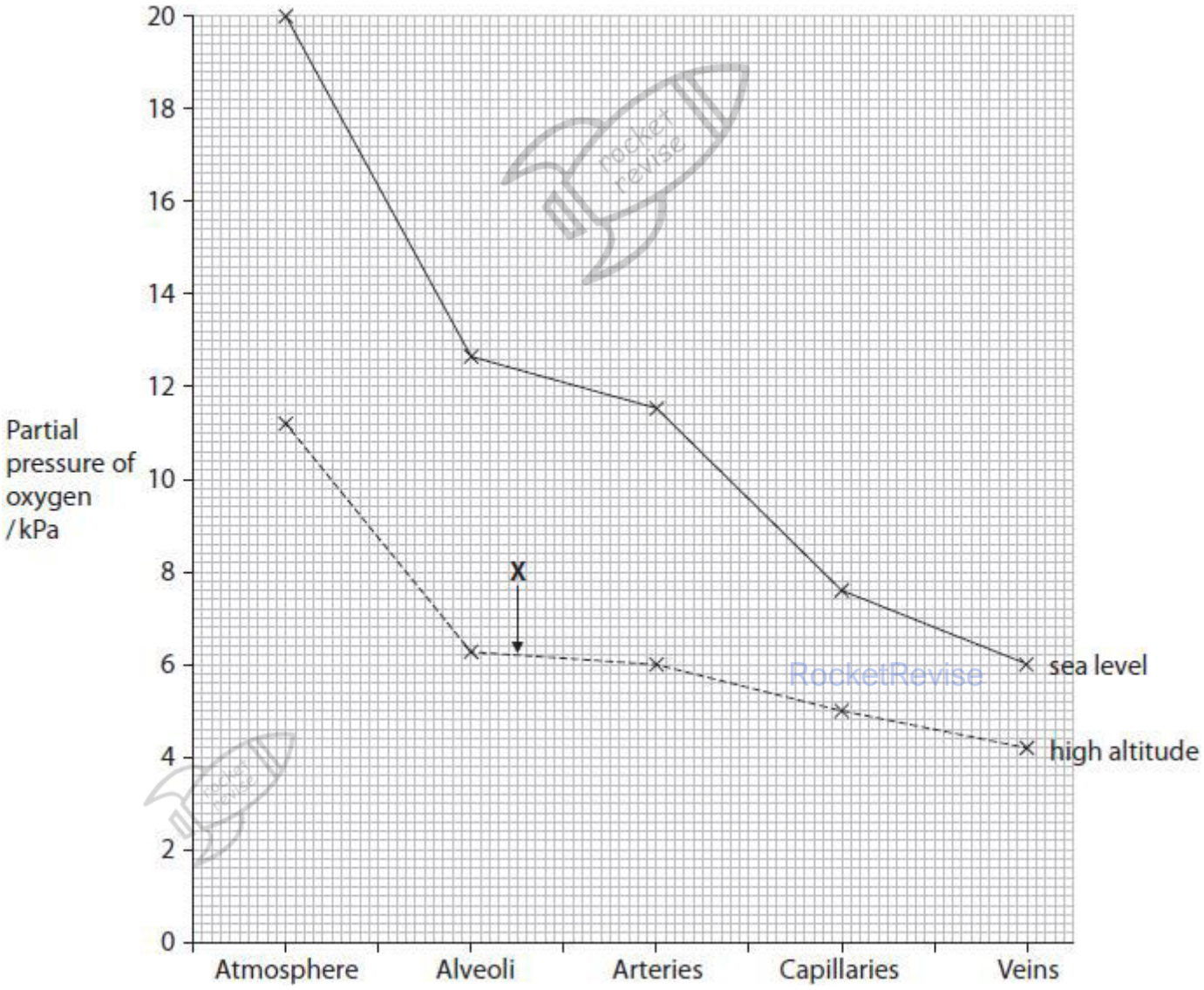
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(b) The oxygen cascade describes the differences in the partial pressure of oxygen between different blood vessels and the external atmosphere.
The graph shows the partial pressures of oxygen for a person at sea level and a person at high altitude.



(i) Suggest why the partial pressure of oxygen in the air in the alveoli is lower than in the atmosphere.

(1)

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(ii) Explain why the partial pressure of oxygen decreases as the blood flows through the arteries and into the veins.

(3)

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(iii) Determine the percentage saturation of haemoglobin in the blood as it leaves the lungs of the person at high altitude.

Use the point marked **X** on the graph and the oxygen dissociation curve of haemoglobin.

(1)

Answer ..... %

(iv) Explain why the percentage saturation of haemoglobin in a person at high altitude is much lower than in a person at sea level.

(4)

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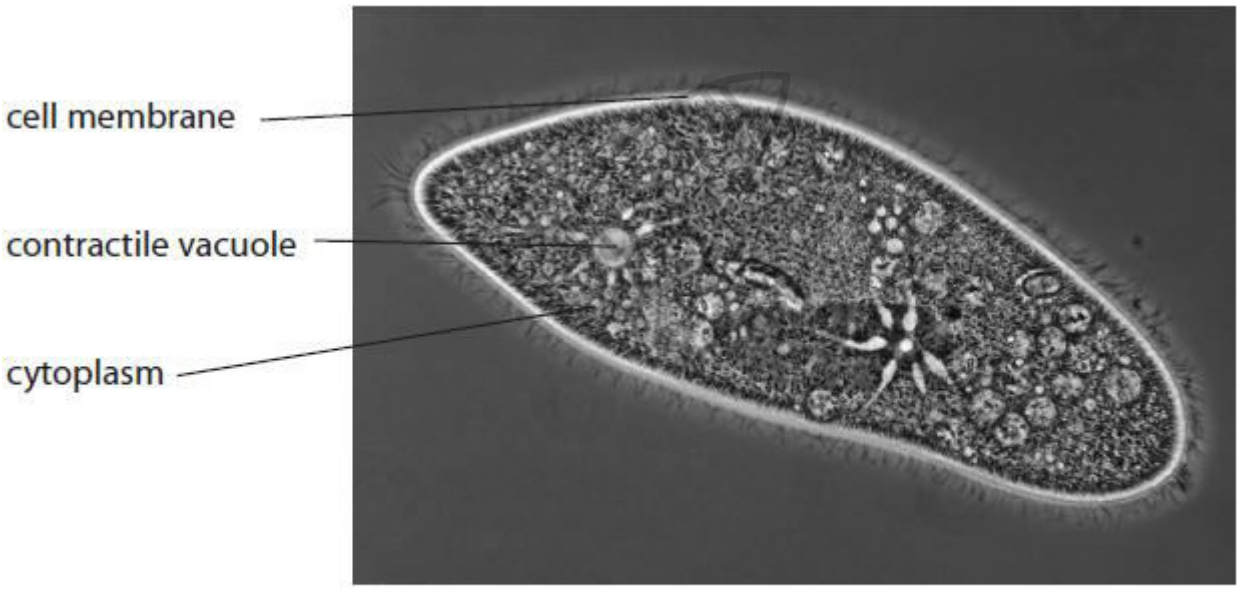
(Total for question = 13 marks)

Q7.

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Paramecia are single-celled organisms that live in pond water.

The photograph shows a single paramecium as seen using a light microscope.



magnification ×400

© M.I. WALKER/SCIENCE PHOTO LIBRARY

Water enters the paramecium. The contractile vacuole pumps water back out of the paramecium.

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(a) (i) Explain why water enters the paramecium.

(2)

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(ii) Explain the importance of the contractile vacuole in the paramecium.

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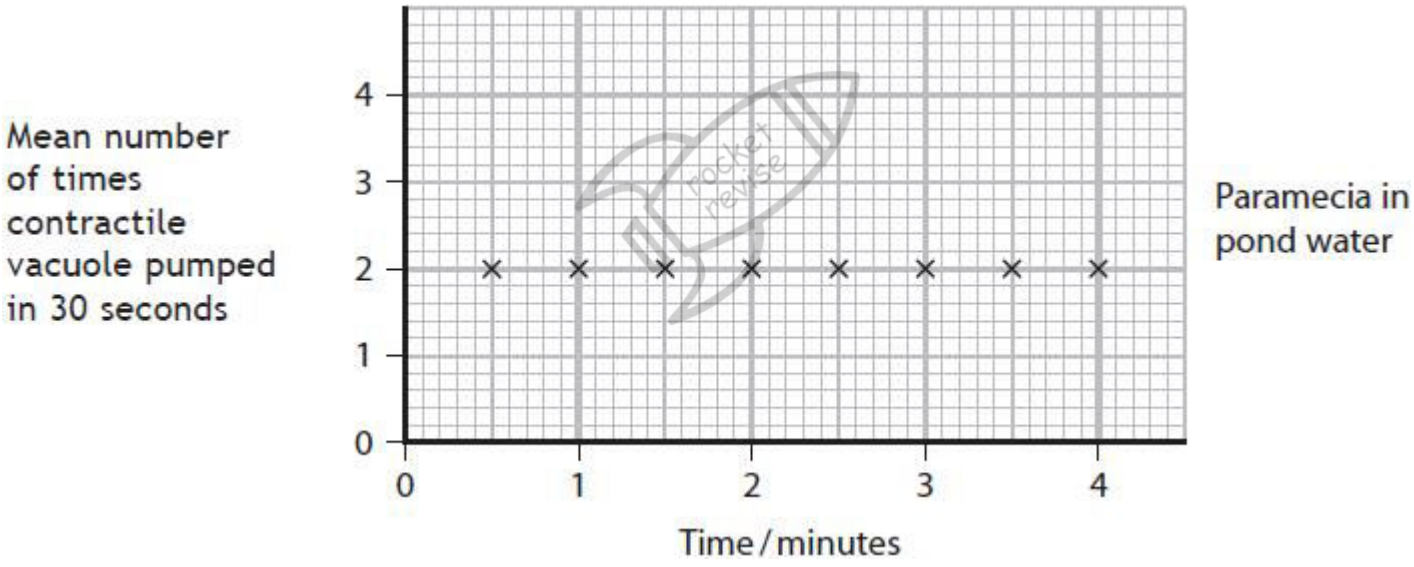
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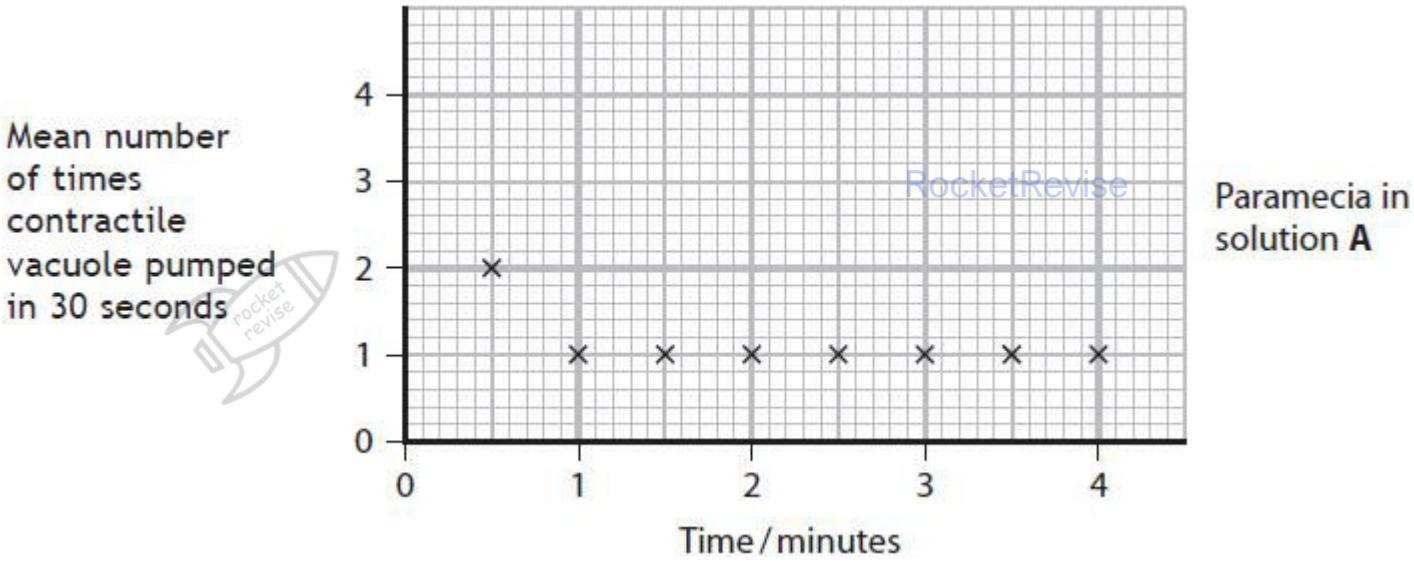
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* (b) In an investigation, paramecia were placed in three separate solutions: pond water, solution **A** and solution **B**.

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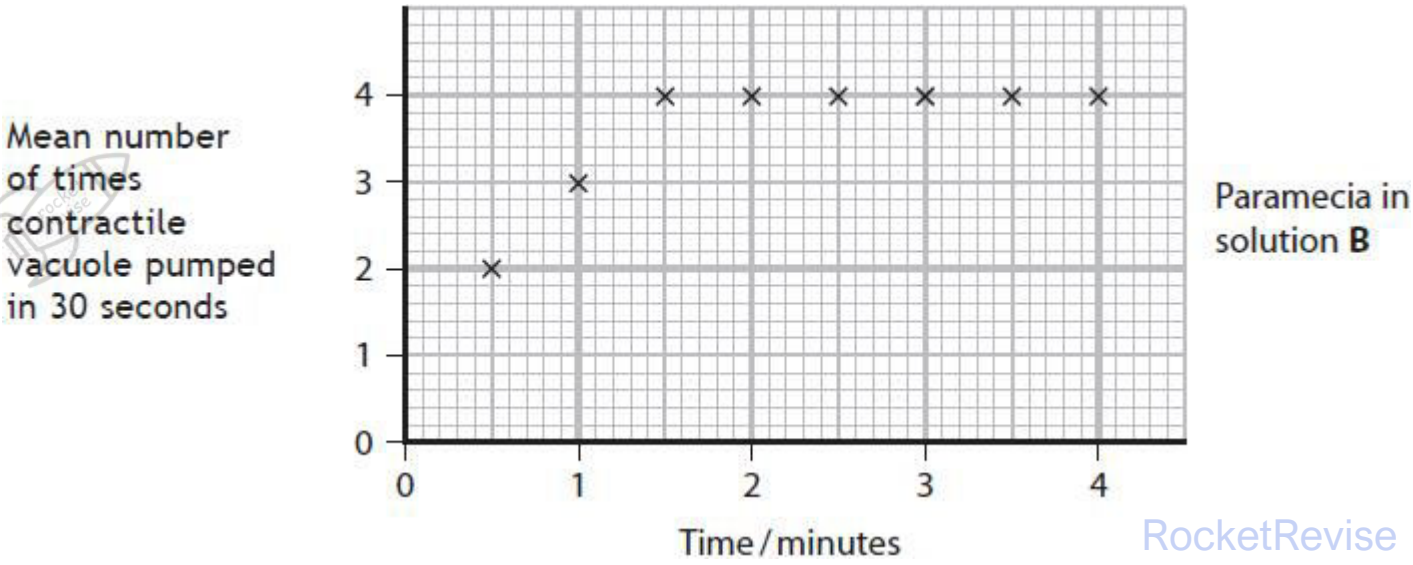
The paramecia were observed using a light microscope.
The number of times that the contractile vacuole pumped water out of the cell in 30-second intervals was recorded.
The graphs show the results of this investigation.





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Explain the results of this investigation.
Use the information in the graphs to support your answer.

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(Total for question = 11 marks)

Q8.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

(a) (i) Which row of the table shows which of these sugars are monosaccharides and which are disaccharides?

(1)

	Monosaccharides	Disaccharides
☐ **A**	fructose only	glucose and sucrose
☐ **B**	glucose only	fructose and sucrose
☐ **C**	fructose and glucose	sucrose only
☐ **D**	glucose and sucrose	fructose only

(ii) Which type of bond joins two monosaccharides together to form a disaccharide?

(1)

☐ **A** ester
☐ **B** glycosidic
☐ **C** peptide
☐ **D** phosphodiester

(b) The table shows the concentration of these sugars in three pineapples.

Sugar	Concentration of sugar / g cm^{-3}		
	Pineapple 1	Pineapple 2	Pineapple 3
fructose	1.71	1.44	1.41
glucose	1.22	1.02	1.00
sucrose	9.08	7.77	8.81

(i) Calculate the mean concentration of glucose in these three pineapples.
Give your answer in **g dm^{-3}**.

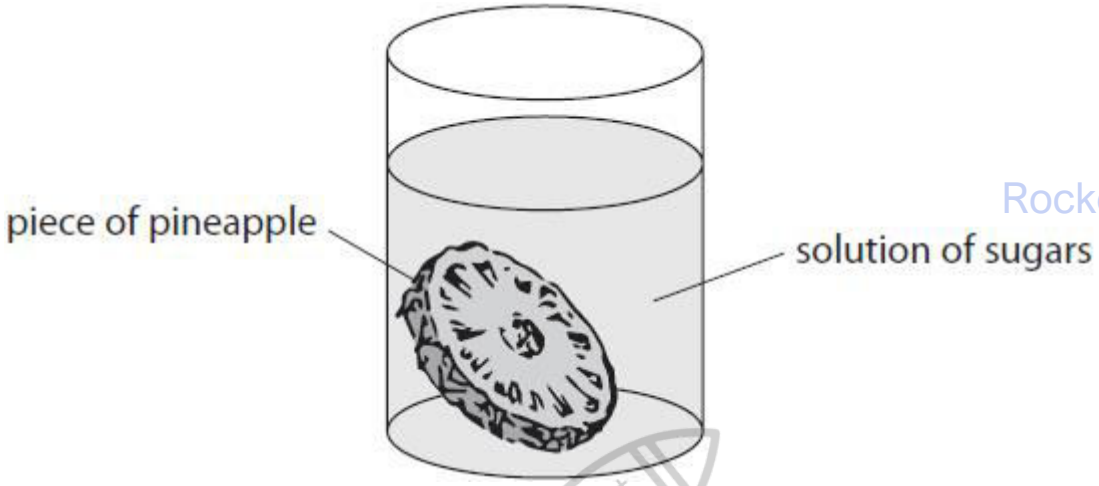
(2)

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Answer ..... g dm^{-3}

* (ii) Pineapples can be preserved by a process called osmotic dehydration in which pieces of pineapple are submerged in a solution of sugars.
The diagram shows how this process can be carried out.



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Osmotic dehydration helps to preserve the pineapple by reducing the water content.
Osmotic dehydration also ensures that the concentration of each sugar in the pineapple does not change. This preserves the sweet taste of the pineapple.
Explain what the solution of sugars should contain to preserve pineapples.
Use the information in the table to support your answer.

(6)

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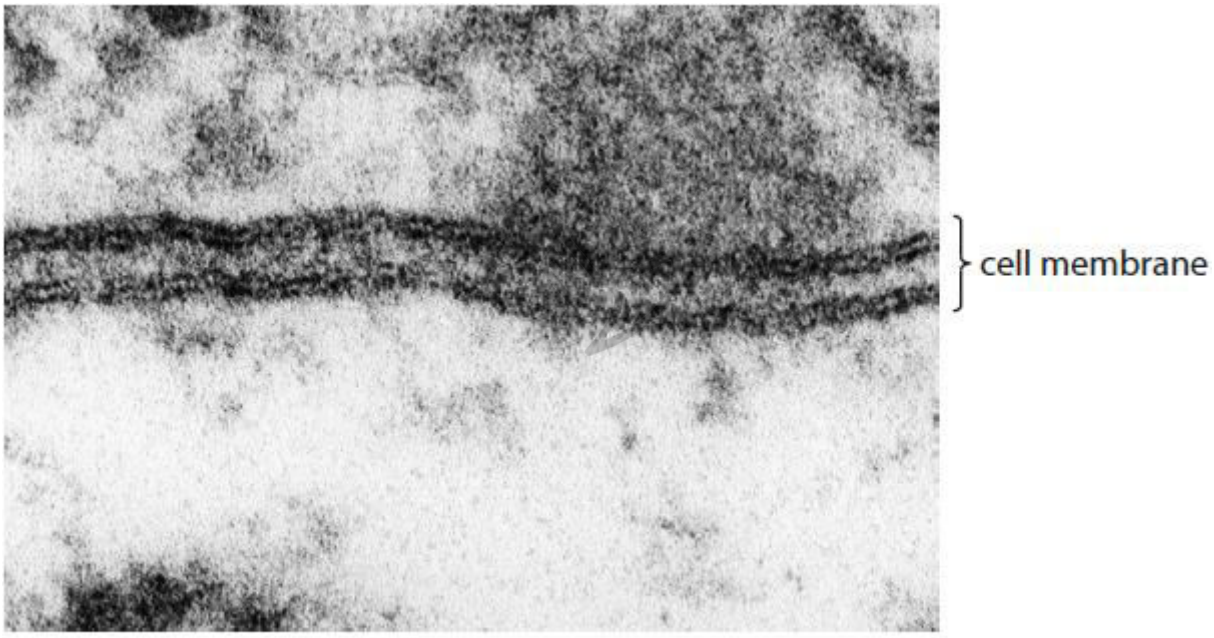
(Total for question = 10 marks)

Q9.

The structure and properties of the cell membrane are important in controlling which molecules can enter and leave a cell.

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(a) The photograph shows part of a cell membrane, as seen using an electron microscope.



(Source: © Dennis Kunkel Microscopy / Science Photo Library)

The width of this cell membrane is 5.00 nm.
A phosphate head of a phospholipid is between 0.8 and 0.9 nm in diameter and the fatty acid tails are between 1.25 and 1.75 nm long.

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Explain how this electron micrograph provides evidence for the structure of the cell membrane.
Use the information given to support your answer.

(3)

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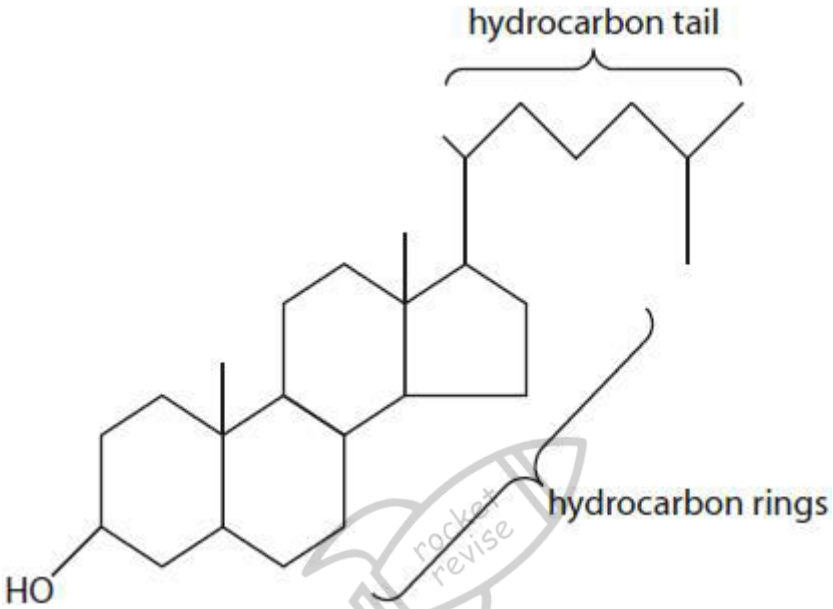
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(b) Cell membranes contain cholesterol.

The molecular formula of cholesterol is $C_{27}H_{46}O$.
The diagram shows a cholesterol molecule.



Explain the location of cholesterol in cell membranes.
Use the information in the diagram to support your answer.

(3)

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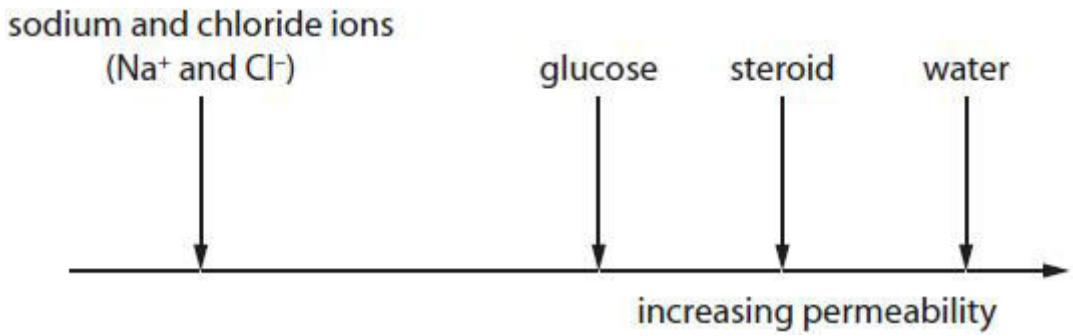
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(c) The diagram shows the permeability of cell membranes to some chemicals.

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The sodium and chloride ions, glucose and water are polar chemicals.
A steroid is a non-polar chemical.

(i) Describe the dipolar nature of water.

(2)

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(ii) Explain the permeability of cell membranes to each chemical shown in the diagram.

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(Total for question = 12 marks)

Q10.

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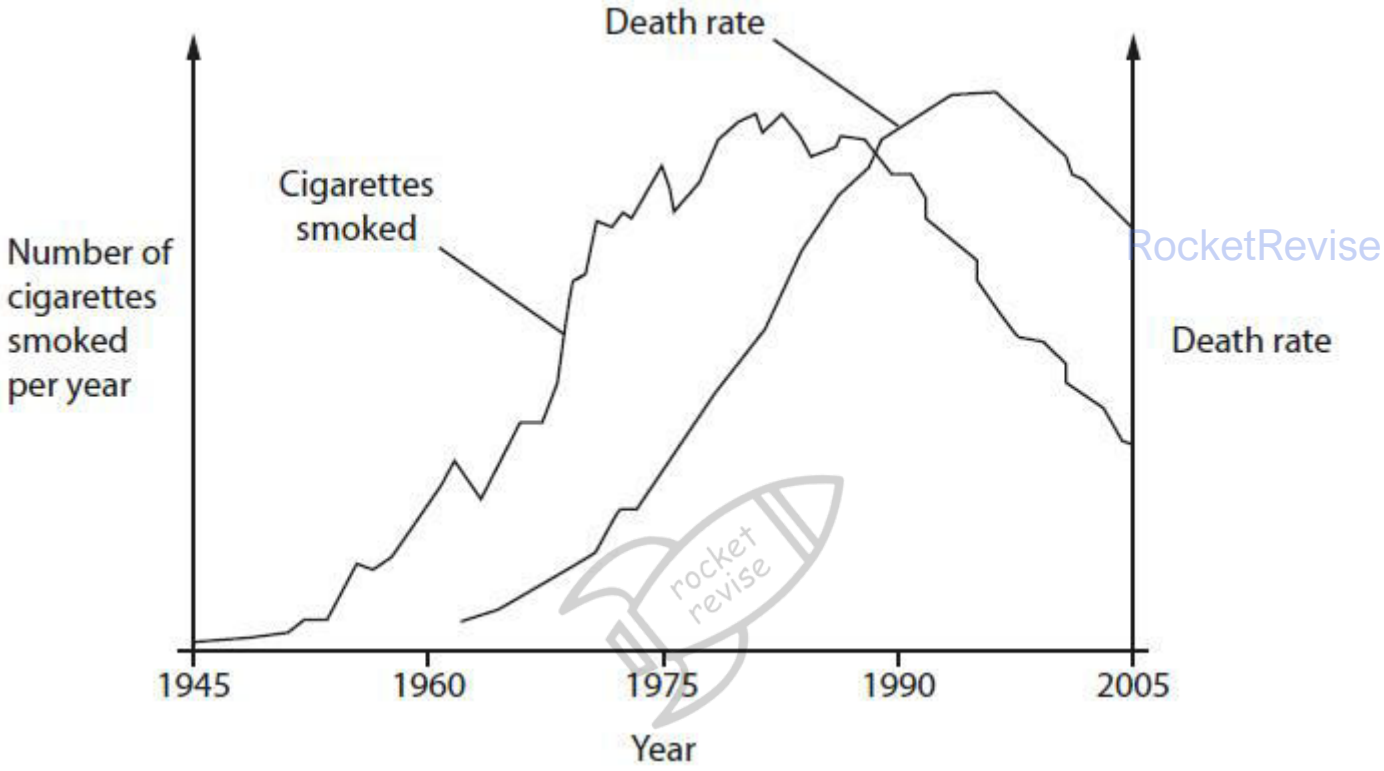
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Gas exchange surfaces have specific adaptations.

(a) Lungs contain the gas exchange surfaces of humans.

Smoking causes lung damage.

(i) The graph shows that there is a correlation between smoking and death rate from lung cancer in men.



Explain how this graph shows that there is a correlation between smoking and death rate from lung cancer in men.

(2)

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(ii) Smoking is a cause of emphysema.

People with emphysema have weakened alveoli that can collapse, creating fewer but larger alveoli.

Explain how this will affect gas exchange in people with emphysema.

(2)

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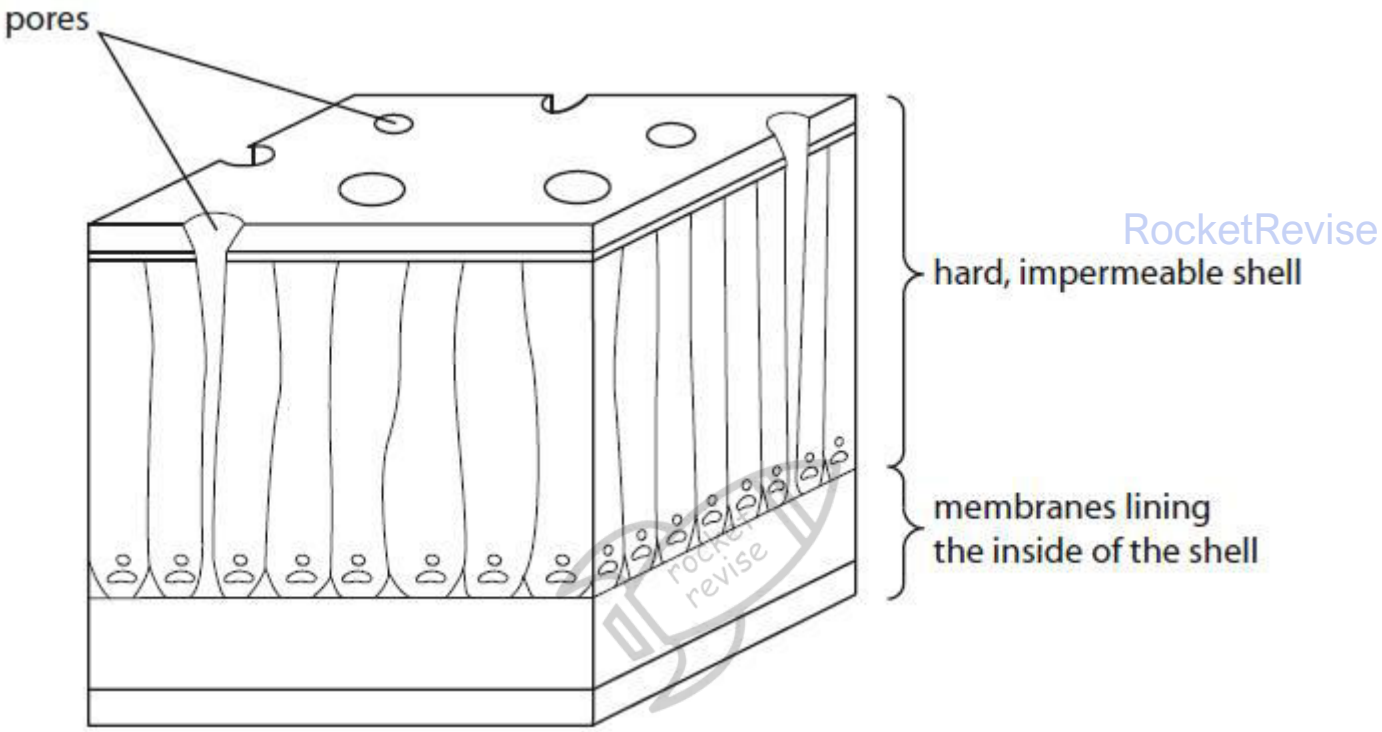
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(b) Bird embryos develop inside hard-shelled eggs.

Gas exchange occurs across the shell of the egg. The oxygen diffuses into the bloodstream of the developing embryo and carbon dioxide diffuses back out.
The diagram shows a section through the shell of an egg.



The thickness of the egg shell is 0.5 mm. The density of the pores in the shell varies from 40 to 400 per cm^2.
Explain the factors that would determine the rate of diffusion of gases between the air and the tissues of the embryo.
Use the information in the diagram, the question and your own knowledge to support your answer.

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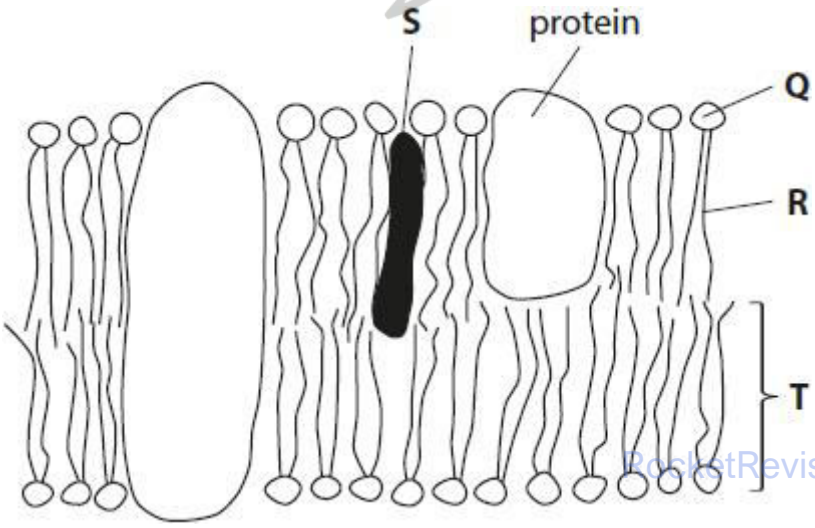
(Total for question = 10 marks)

Q11.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The structure of the cell membrane affects the properties of the cell membrane.

The diagram shows the structure of part of a cell membrane.



(a) (i) The magnification of this diagram is 1×10^7.

The structure labelled **T** is 2 units long.
What is this unit?

(1)

- ☒ **A** cm
- ☒ **B** mm
- ☒ **C** µm
- ☒ **D** nm

(ii) Which structures contain a phosphate group?

(1)

- ☒ **A** **Q** and **R**

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- ☐ **B** **Q** and **T**
- ☐ **C** **R** and **S**
- ☐ **D** **S** and **T**

(iii) Which ratio of the structures affects the fluidity of this membrane?

(1)

- ☐ **A** **Q** : **T**
- ☐ **B** **R** : **Q**
- ☐ **C** **R** : **T**
- ☐ **D** **S** : **T**

(b) Explain the role of the primary structure in determining the properties of the protein labelled in the diagram.

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*(c) The diagram and table give information about some molecules found inside and outside a cell.

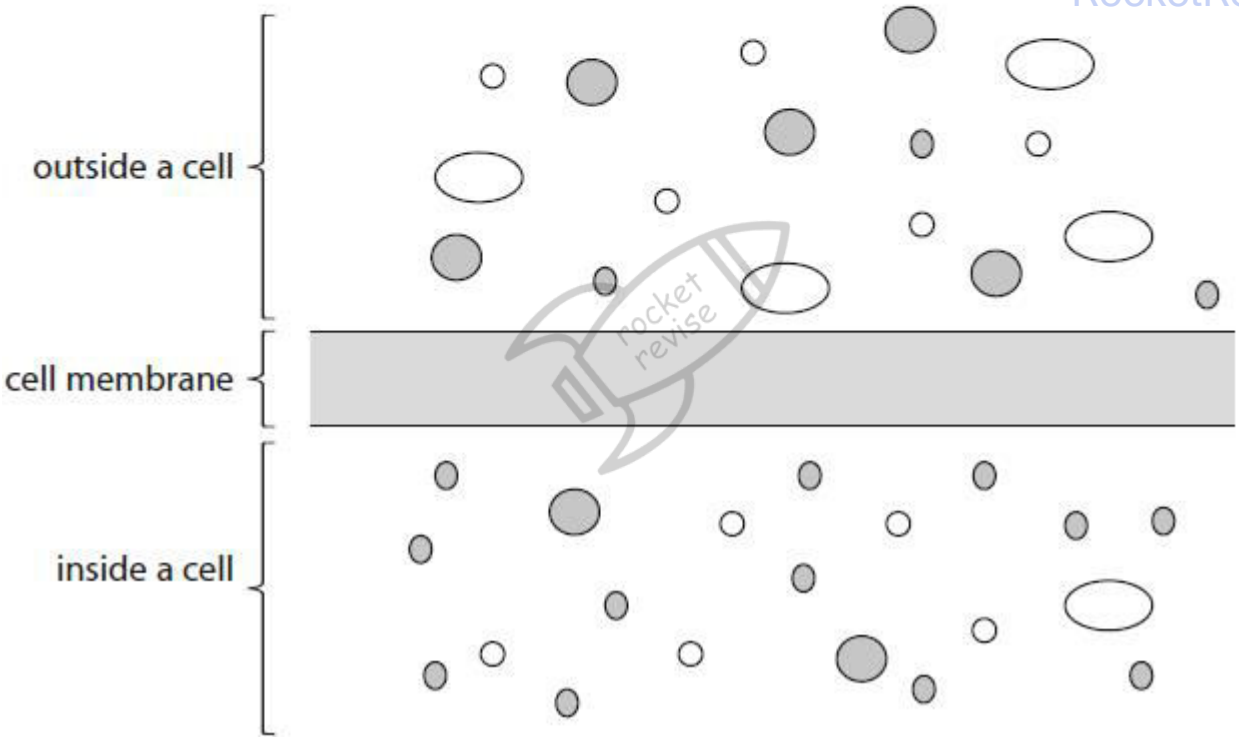
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Molecule	Key	Description
E	○	small dipolar molecule consisting of two hydrogen atoms and one oxygen atom
F	●	large polar molecule
G	⬭	non-polar molecule
H	●	small polar molecule

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Explain why each of these molecules **enters** the cell by a different mechanism.
Use the information in the table and the diagram to support your answer.

(6)

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(Total for question = 13 marks)

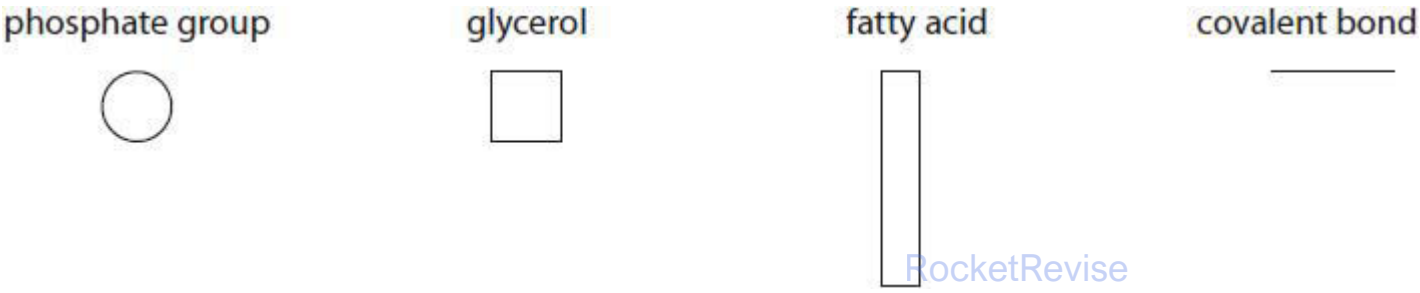
Q12.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The cell membrane controls which substances can enter and leave a cell.

The cell membrane is a phospholipid bilayer with proteins embedded in it.

(a) (i) The shapes show the components of a phospholipid.



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Draw a diagram to show the structure of a phospholipid, using these shapes.

(2)

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(ii) Which type of bond is found in this molecule?

(1)

- ☒ **A** ester
- ☒ **B** glycosidic
- ☒ **C** hydrogen
- ☒ **D** peptide

(b) (i) How does a nonpolar molecule move through a cell membrane, down a concentration gradient into a cell?

(1)

- ☒ **A** by active transport
- ☒ **B** by diffusion
- ☒ **C** by exocytosis
- ☒ **D** by osmosis

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(ii) How many of the following statements are correct for the transport of a substance by endocytosis?

- transports substances out of the cell
- the cell membrane surrounds the substance
- can only transport substances down a concentration gradient (1)

- ☒ **A** none
- ☒ **B** one
- ☒ **C** two
- ☒ **D** three

(Total for question = 5 marks)

Q13.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The components of a cell membrane determine its properties.

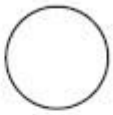
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The membrane is a phospholipid bilayer.

(a) The table shows one way that the components of phospholipids can be drawn.

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Component	Way of drawing the component
phosphate group	
glycerol	
fatty acid	
bond	

(i) Draw a diagram of a phospholipid, using the information in the table.

(2)

(ii) One of the bonds present in phospholipids is an ester bond.

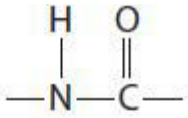
Which diagram shows an ester bond?

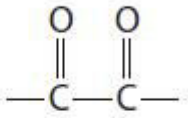
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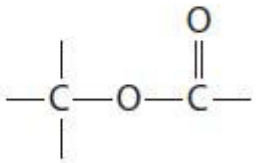
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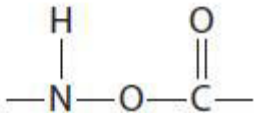
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☒ **A** 

☒ **B** 

☒ **C** 

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☒ **D** 

(b) The fluidity of a membrane is increased by a number of factors:

- increase in temperature
- increase in the proportion of unsaturated fatty acids
- decrease in cholesterol
- fatty acids with shorter side chains.

(i) Explain why an increase in temperature increases the fluidity of the membrane.

(2)

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(ii) Which row in the table describes an unsaturated fatty acid compared with a saturated fatty acid with the same number of carbon atoms?

(1)

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	Has double bonds between two carbons	Number of hydrogens
☒ A	no	fewer than the saturated fatty acid
☒ B	no	more than the saturated fatty acid
☒ C	yes	fewer than the saturated fatty acid
☒ D	yes	more than the saturated fatty acid

(iii) Explain why a decrease in cholesterol increases the fluidity of the membrane.

(2)

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(iv) Suggest why a fatty acid with a shorter side chain will increase the fluidity of a membrane.

(1)

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(c) Studies have shown that when fish are moved into water at a lower temperature, the fatty acid content of their membranes changes.

(i) Suggest how the content of the fatty acids in these membranes may have changed.

(2)

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(ii) Suggest why these changes are necessary for the survival of the fish.

(2)

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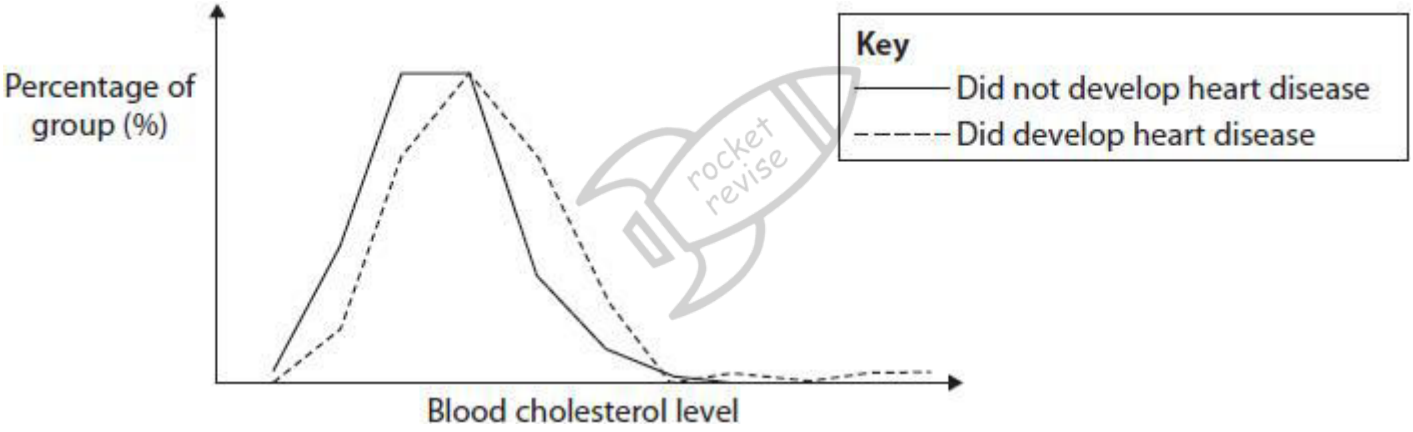
(Total for question = 13 marks)

Q14.

A number of factors increase the risk of cardiovascular disease (CVD), including blood cholesterol levels.

(a) In one investigation, scientists measured the blood cholesterol level in a group of people and monitored the development of heart disease in the following 10 years.

The graph shows the results of this investigation.



Some people interpreted the data as showing evidence for a relationship between blood cholesterol levels and heart disease.

Other people interpreted the data as not showing evidence for a relationship between blood cholesterol levels and heart disease.

Describe the evidence for and against a relationship between blood cholesterol levels and heart disease.

(2)

Evidence for

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Evidence against

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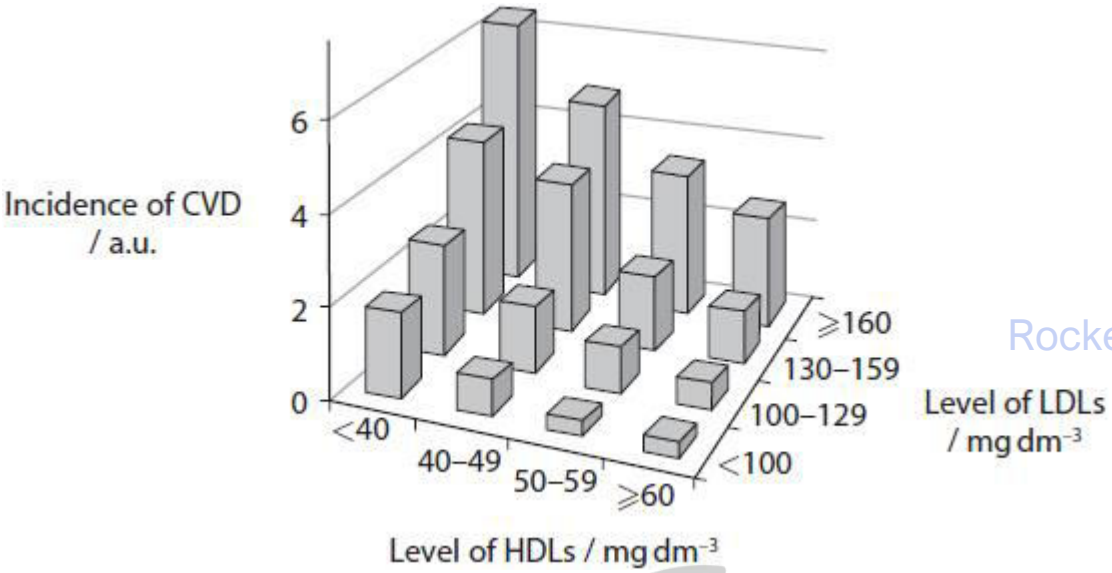
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(b) Other investigations have shown that the levels of high-density lipoproteins (HDLs) and low-density lipoproteins (LDLs) are significant in determining the risk of CVD.

The graph shows the relationship between levels of HDLs and LDLs on the incidence of CVD.



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The following conclusions can be made from this graph:

- as the levels of LDLs increases so does the incidence of CVD
- as the levels of HDLs increases the incidence of CVD decreases
- the higher the ratio of HDLs to LDLs the lower the incidence of CVD.

Describe the evidence shown in this graph that support these conclusions.

(3)

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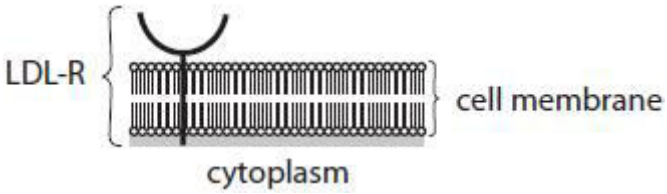
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(c) The low-density lipoprotein receptor (LDL-R) is a protein on the surface of liver cells that is involved in the endocytosis of LDLs.

The diagram shows an LDL-R in a cell membrane.

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(i) Explain how the LDL-R is involved in endocytosis of LDLs.

(3)

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(ii) The lower pH inside the cell causes the LDL to separate from its receptor. Suggest how a change in pH could cause this separation.

(2)

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(iii) Mutations in the gene coding for the LDL-R can result in an increased risk of CVD. Explain why mutations in this gene can increase the risk of CVD.

(3)

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(Total for question = 13 marks)

Q15.

Some questions must be answered with a cross in a box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

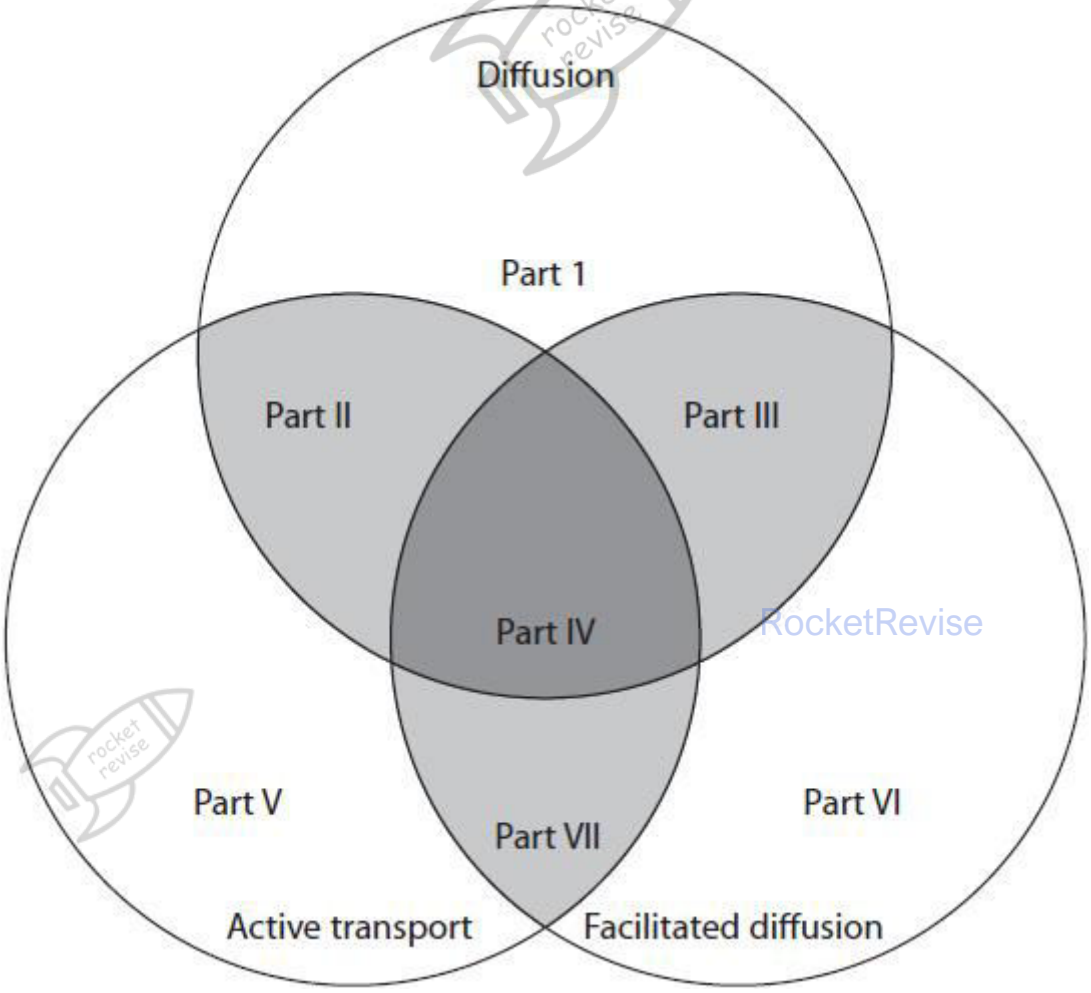
The cell membrane controls which substances can move into the cell or out of the cell.

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Processes by which substances can move into a cell or out of a cell include diffusion, facilitated diffusion, active transport and osmosis.

(a) A Venn diagram can be used to show the similarities and differences between diffusion, facilitated diffusion and active transport.

Statements about similarities can be written in the numbered parts of the circles that overlap.
For example, statements about similarities shared by all three processes would be written in Part IV.
Statements about differences can be written in the numbered parts of the circles that do not overlap.



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(i) Which part would contain the statement: uses proteins?

☒ **A** II
☒ **B** III

(1)

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- **C** IV
- **D** VII

(ii) Which part would contain the statement: needs energy in the form of ATP?

(1)

- **A** II
- **B** V
- **C** VI
- **D** VII

(iii) Which part would contain the statement: solutes can only move down a concentration gradient?

(1)

- **A** II
- **B** III
- **C** V
- **D** VII

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(b) Osmosis can be defined as the movement of free water molecules through a partially permeable membrane, down a water potential gradient.

Explain this definition.

(3)

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*(c) The photograph shows a Chinese mitten crab.



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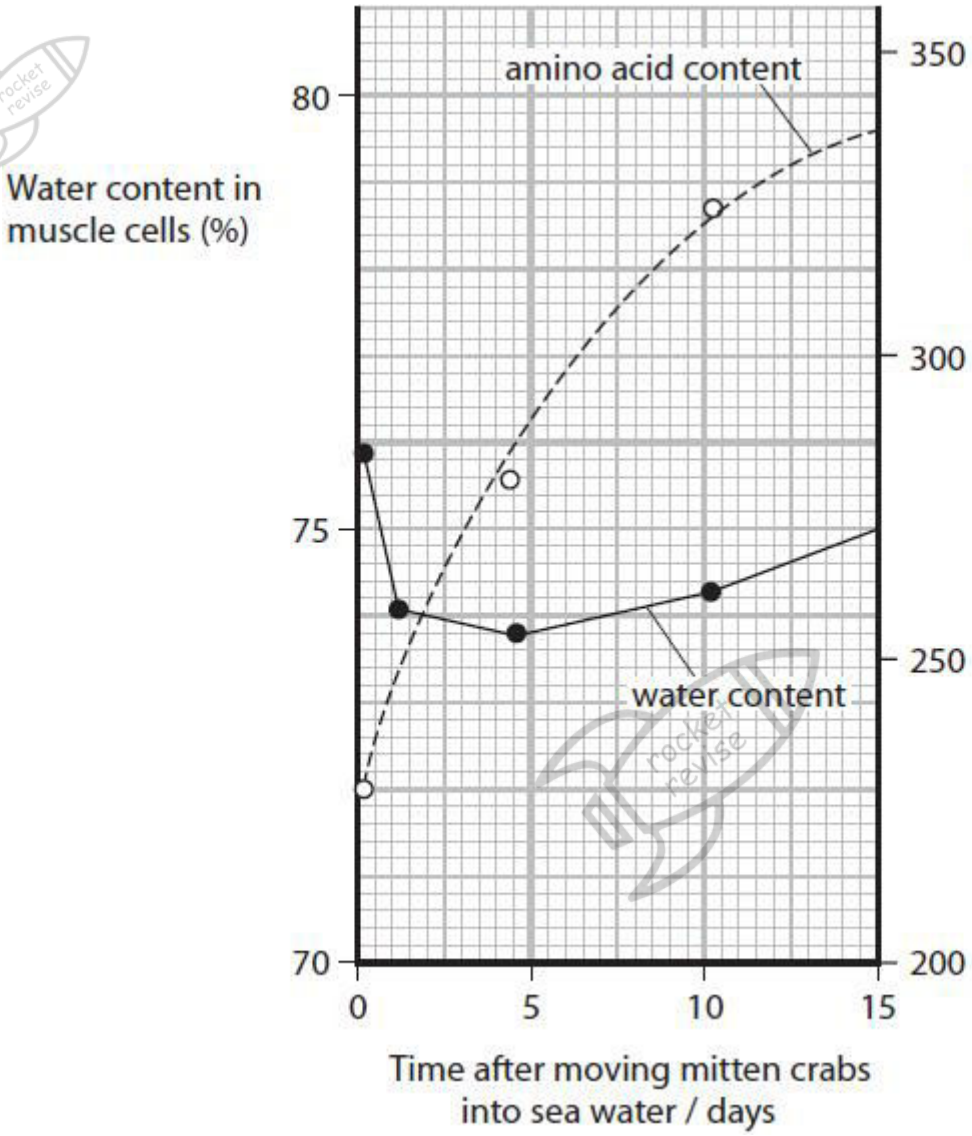
(Source: © WILDLIFE GmbH / Alamy Stock Photo)

Mitten crabs spend most of their lives in fresh water and only return to the sea to breed.
In an investigation, mitten crabs were kept in fresh water and then moved into sea water.

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The water content and amino acid content in the muscle cells of these crabs were measured for 15 days after moving them from fresh water into sea water.
The graph shows the results of this investigation.



Sea water has a higher concentration of salt than fresh water.
Explain the changes in water content and amino acid content in the muscle cells of the crabs in this investigation.

(6)

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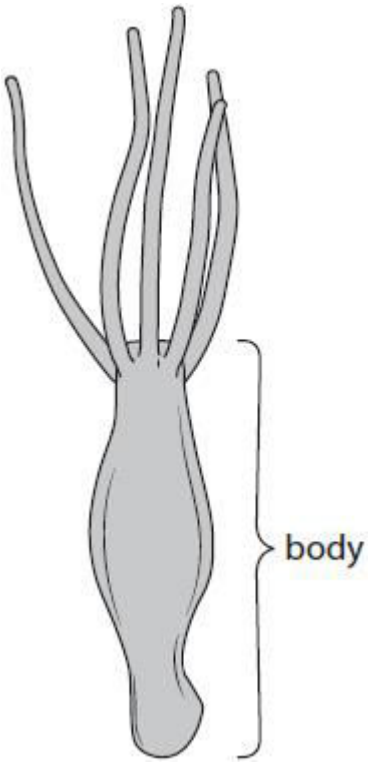
(Total for question = 12 marks)

Q16.

Hydra are small animals that live in water.

They have a cylindrical body.

The drawing shows a hydra.



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(a) This hydra has a body length of 1.8 mm and a diameter of 1 mm.

(i) Which is the magnification of this drawing?

(1)

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- ☐ **A** 3.75×10^{-2}
- ☐ **B** 3.75×10^{-1}
- ☐ **C** 2.67×10^{0}
- ☐ **D** 2.67×10^{1}

(ii) Which is the surface area, in mm^2, of this hydra?
Use the formula:
surface area = 2ℼrl

(1)

- ☐ **A** 0.9 ℼ
- ☐ **B** 1.8 ℼ
- ☐ **C** 3.6 ℼ
- ☐ **D** 5.4 ℼ

(iii) Another hydra has a surface area of 2.3 ℼ mm^2, a body length of 1.8 mm and a diameter of 1.2 mm.
Calculate the surface area to volume ratio of this hydra.
Use the formula:
volume = ℼr^2l

(2)

Answer .....

(b) A hydra has a hollow body that is made up of two layers of cells.

Explain how a hydra is adapted for gas exchange.

(3)

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(Total for question = 7 marks)

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Q17.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Gas exchange requires diffusion.

The rate of diffusion is dependent on the surface area, the diffusion distance and the concentration gradient.

The rate of diffusion can be calculated using the formula:

$$\text{rate of diffusion} \propto \frac{\text{surface area} \times \text{concentration gradient}}{\text{diffusion distance}}$$

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(a) Name the law on which this formula is based.

(1)

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(b) What will happen to the rate of diffusion if all three of the following are changed at the same time?

- the surface area is halved
- the concentration gradient is halved
- the diffusion distance is halved

(1)

☒ **A** it will stay the same
☒ **B** it will be a quarter of its value
☒ **C** it will halve
☒ **D** it will double

(c) The properties of gas exchange surfaces include:

- a large surface area
- a short diffusion distance
- a high concentration gradient.

Describe how mammalian lungs are adapted to have each of these three properties.

(3)

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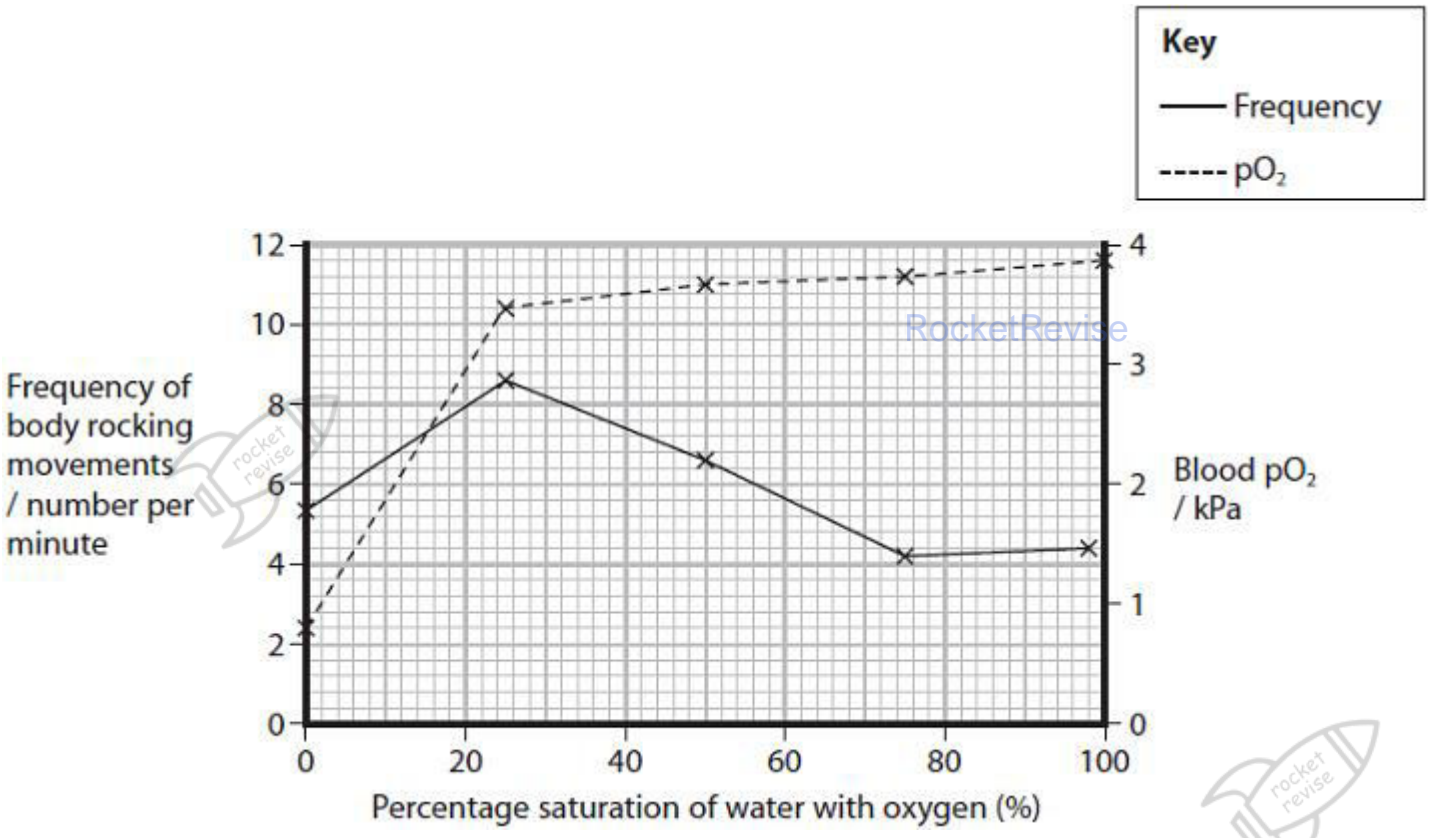
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*(d) The photograph shows a hellbender salamander.



(Source: © Robert Hamilton/Alamy Stock Photo)

These salamanders live in freshwater streams that are shallow and fast-flowing.
They use their lungs and the surface of their skin for gas exchange.
They also use body rocking and swaying movements in the water.
The graph shows the relationship between the frequency of body rocking movements, blood pO_2 and the percentage saturation with oxygen of the water they live in.



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Discuss how the structure and behaviour of these salamanders are adapted for gas exchange.

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(Total for question = 11 marks)

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Q18.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Some marine animals are osmoconformers and some are osmoregulators.

Strict osmoconformers have a cell solute concentration that matches the solute concentration of the sea water.

Partial osmoconformers have a cell solute concentration that varies with the solute concentration of the sea water.

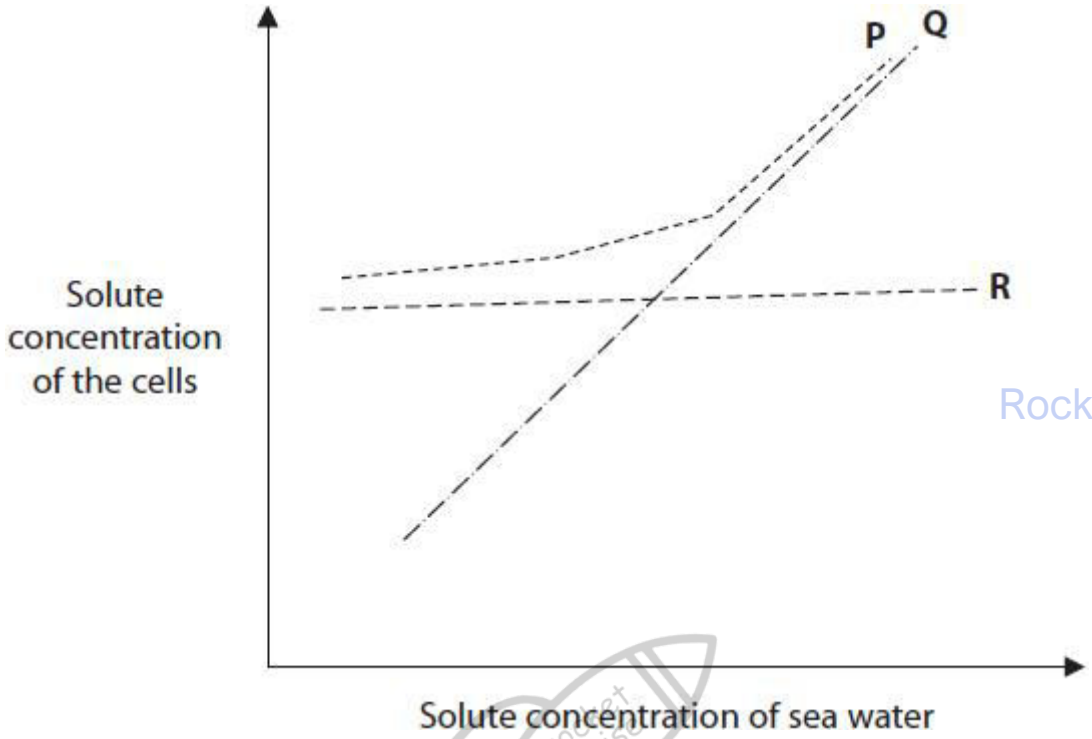
Osmoregulators control the solute concentration of their cells when the solute concentration of sea water

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changes.

(a) The graph shows the solute concentration of the cells of three marine animals, P, Q and R.



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(i) Which row in the table identifies the marine animals P, Q and R?

(1)

		P	Q	R
☒	**A**	partial osmoconformer	osmoregulator	strict osmoconformer
☒	**B**	partial osmoconformer	strict osmoconformer	osmoregulator
☒	**C**	osmoregulator	strict osmoconformer	partial osmoconformer
☒	**D**	strict osmoconformer	partial osmoconformer	osmoregulator

(ii) Explain the advantages for osmoconformers of having cells with solute concentrations similar to that of the sea water they live in.

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(3)

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(b) The table shows the concentration of ions in sea water and in some osmoconformers.

Location		Ion concentration / a.u.				
		sodium	magnesium	calcium	potassium	chloride
Sea water		478	55	11	10	558
Cells of osmoconformers	jellyfish	474	53	10	11	
	sea urchin	474	54	11	10	557
	lobster	541	9	12	8	552

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(i) Estimate the chloride concentration of the cells of a jellyfish.

(1)

Answer ..... a.u.

(ii) Calculate the percentage difference in the concentration of magnesium ions in the lobster compared with the sea urchin.

Give your answer to two decimal places.

(2)

Answer ..... %

(iii) These three animals are osmoconformers even though they have different concentrations of each ion in their cells.

Explain why these three animals can be described as osmoconformers.

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(2)

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(iv) Suggest why osmoconformers do not have the same concentration of each ion in their cells.

(1)

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(v) Energy is needed to maintain the concentration of ions inside the osmoconformers. Explain what this energy is being used for.

(2)

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(Total for question = 12 marks)

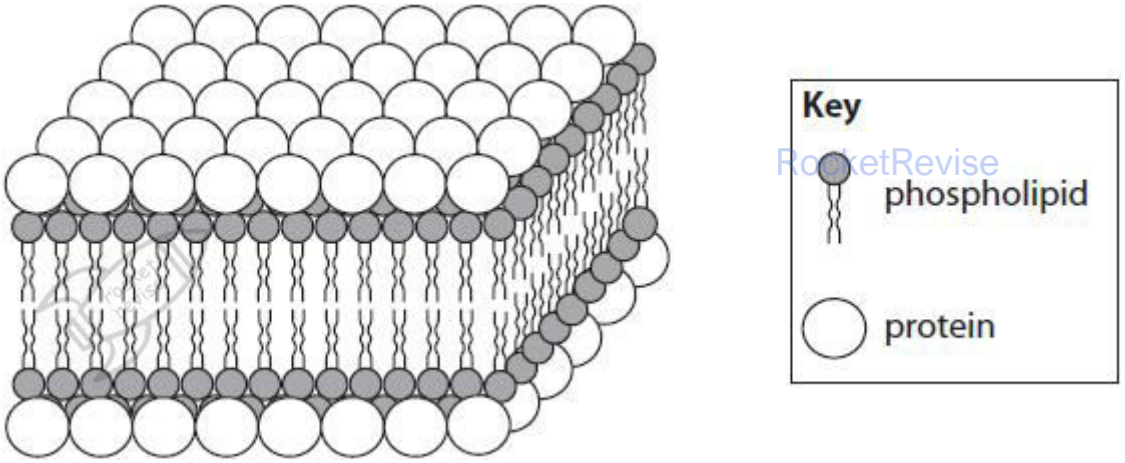
Q19.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The fluid mosaic model of membrane structure was developed to explain the structure and properties of cell membranes.

(a) One model of membrane structure was developed by Davson and Danielli.

The diagram shows the Davson–Danielli model.



(i) Which describes a phospholipid?

(1)

- ☒ **A** hydrophilic fatty acid head and hydrophobic phosphate tails
- ☒ **B** hydrophilic phosphate head and hydrophobic fatty acid tails
- ☒ **C** hydrophobic fatty acid head and hydrophilic phosphate tails
- ☒ **D** hydrophobic phosphate head and hydrophilic fatty acid tails

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(ii) Compare and contrast the structure of the model shown in the diagram with that of the fluid mosaic model of membrane structure.

(4)

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(b) The cell membrane is important to the cell and if it gets damaged the cell could die.

(i) Suggest how damage to the cell membrane could result in cell death.

(2)

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(ii) Cells can repair damage to the cell membrane.
This involves fusion of parts of the membrane around the edge of the damage.
Explain why parts of the membrane are able to fuse together to repair the damage.

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(2)

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(iii) Once the membrane has fused together, other processes are needed to complete the repair of the membrane.

Suggest what else the cell needs to do to complete this repair.

(2)

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(Total for question = 11 marks)

Q20.

The cell membrane is important in the control of which substances can enter and leave the cell.

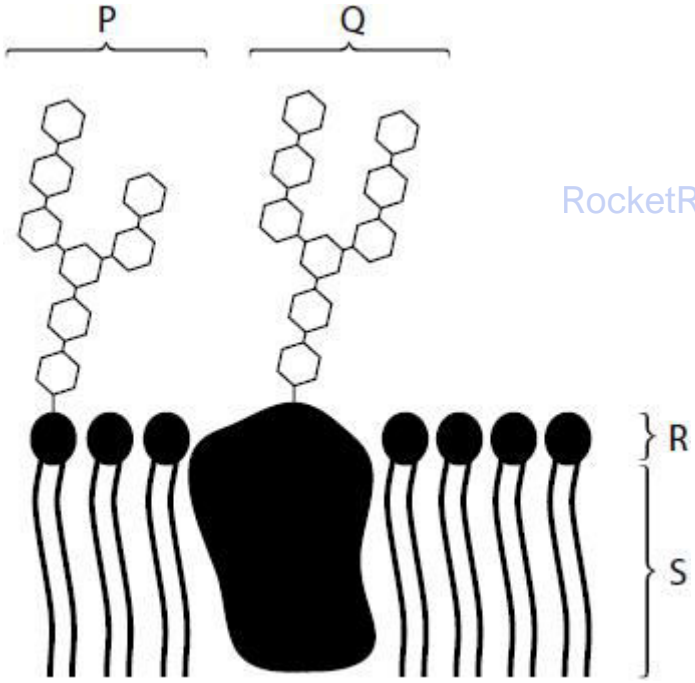
(a) The cell membrane consists of a phospholipid bilayer.

(i) Why do phospholipids form a bilayer?

(1)

- ☒ **A** the hydrophobic heads dissolve in the aqueous (water) environment
- ☒ **B** the hydrophobic heads move away from the aqueous (water) environment
- ☒ **C** the hydrophobic tails dissolve in the aqueous (water) environment
- ☒ **D** the hydrophobic tails move away from the aqueous (water) environment

(ii) This diagram shows part of a cell membrane.



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Which letter represents a membrane glycoprotein?

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(1)

- ☐ **A** P
- ☐ **B** Q
- ☐ **C** R
- ☐ **D** S

(b) State what is meant by the term **osmosis.**

(1)

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(c) Compare and contrast exocytosis and endocytosis.

(3)

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(d) Explain why oxygen molecules can pass directly through the cell membrane.

(3)

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(Total for question = 9 marks)

Q21.

(a) Explain how human lungs are adapted for rapid gas exchange.

(3)

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(b) Cystic fibrosis is an inherited condition which reduces gas exchange.

The most frequently observed CFTR allele associated with cystic fibrosis carries the ΔF508 mutation.

(i) State what is meant by the term **allele**.

(1)

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(ii) The CFTR protein coded for by this mutation has one missing amino acid compared to the functioning protein.

Explain how this mutation results in a non-functioning CFTR protein.

(4)

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(iii) Explain why people with cystic fibrosis may develop lung infections.

(4)

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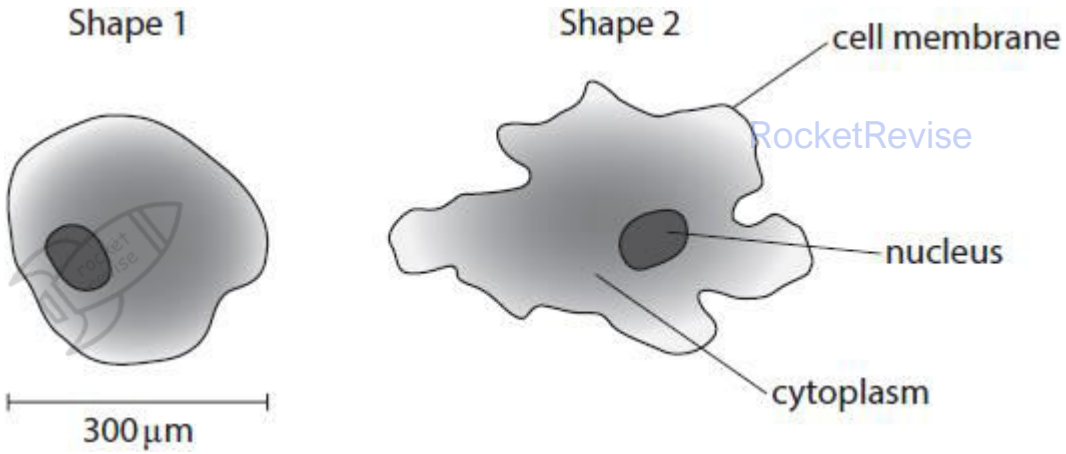
(Total for question = 12 marks)

Q22.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Amoeba are single-celled organisms that live in water. They are able to change their shape.

The diagram shows two different shapes that one amoeba can make.



(a) An amoeba changes shape by pushing cytoplasm against the cell membrane, forming long thin projections.

State the property of the cell membrane that enables the amoeba to change shape.

(1)

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(b) Gas exchange in an amoeba takes place across its cell membrane.

(i) The amoeba can be roughly spherical, as shown in Shape 1.
Which is the surface area of the amoeba in this shape?
Use the formula:

surface area of a sphere = $4\pi r^2$

(1)

- ☒ **A** 90 000π μm^2
- ☒ **B** 90 000π μm^3
- ☒ **C** 360 000π μm^2
- ☒ **D** 360 000π μm^3

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(ii) The mass of the amoeba stays the same when it changes from Shape 1 to Shape 2.
Which row of the table shows what happens to the surface area and volume of this amoeba when it changes to Shape 2?

(1)

	surface area	volume
☒ **A**	decreases	decreases
☒ **B**	increases	increases
☒ **C**	increases	stays the same
☒ **D**	stays the same	stays the same

(iii) Explain why an amoeba can rely on diffusion to supply the oxygen it needs.

(2)

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(Total for question = 5 marks)

Q23.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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The cell membrane determines which molecules can enter or leave a cell.

(a) Which pair of membrane transport mechanisms only involves a **solute** moving down a concentration gradient?

(1)

- ☐ **A** active transport and diffusion
- ☐ **B** active transport and osmosis
- ☐ **C** diffusion and facilitated diffusion
- ☐ **D** facilitated diffusion and osmosis

(b) Which of the following transport mechanisms involve carrier proteins in the membrane?

- active transport
- diffusion
- exocytosis

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(1)

- ☐ **A** active transport only
- ☐ **B** active transport and exocytosis
- ☐ **C** diffusion only
- ☐ **D** diffusion and exocytosis

(c) Which cell transport mechanism is used to take large particles or bacteria into a cell?

(1)

- ☐ **A** active transport
- ☐ **B** endocytosis
- ☐ **C** exocytosis
- ☐ **D** facilitated diffusion

(d) Which cell transport mechanism moves small lipid-soluble molecules through the membrane?

(1)

- ☐ **A** active transport
- ☐ **B** diffusion
- ☐ **C** facilitated diffusion
- ☐ **D** osmosis

(e) Channel proteins are involved in facilitated diffusion.

Which row of the table describes the charge on the R groups on the amino acids of a channel protein?

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(1)

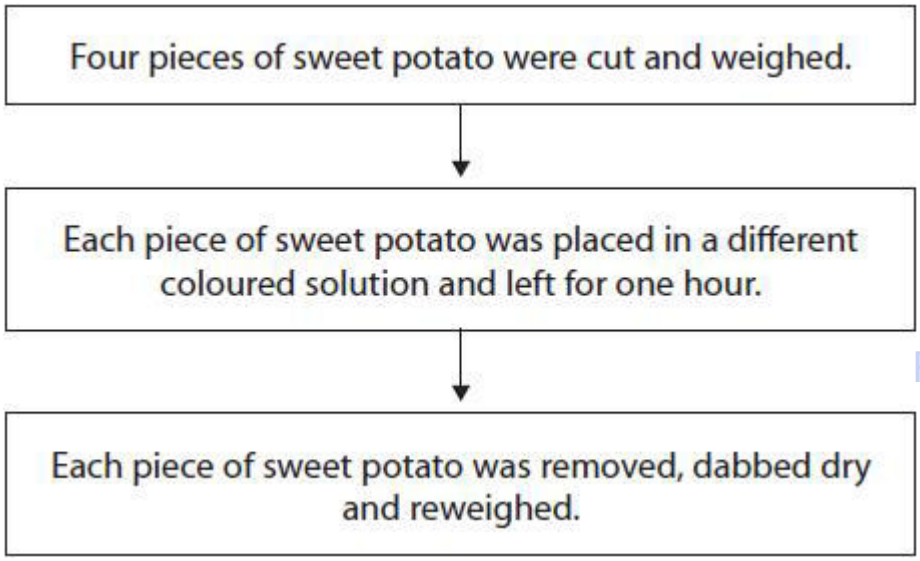
	Charge on R groups facing inside the channel are mostly	Charge on R groups facing the rest of the membrane are mostly
☐ **A**	polar	polar
☐ **B**	polar	non-polar
☐ **C**	non-polar	polar
☐ **D**	non-polar	non-polar

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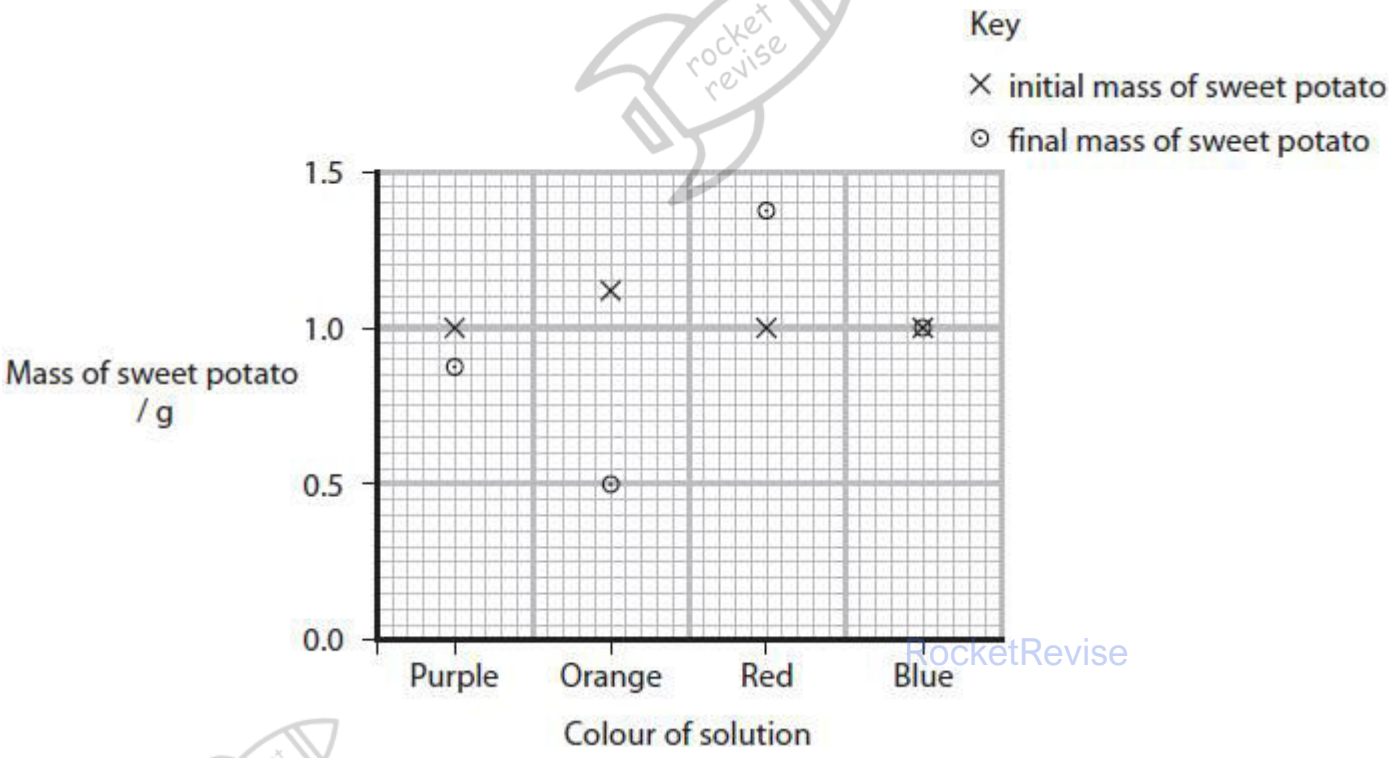
(f) A student investigated the effect of four different coloured solutions on the mass of pieces of sweet potato.

Dyes were dissolved in water to produce the coloured solutions.
The diagram shows the method used by the student.



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The graph shows the results of this investigation.



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Explain the results of this investigation.

(4)

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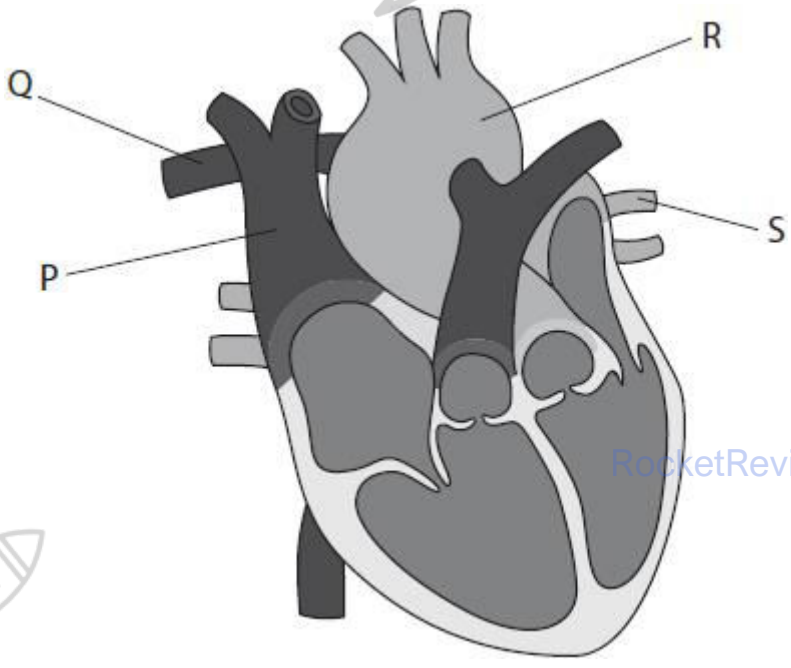
(Total for question = 9 marks)

Q24.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Many animals have a heart and circulation.

(a) The diagram shows a human heart and its major blood vessels, labelled P, Q, R and S.



(i) Which pair of blood vessels transports blood to and from the lungs?

(1)

- ☒ **A** P and Q
- ☒ **B** Q and S
- ☒ **C** R and P
- ☒ **D** S and R

(ii) Which blood vessels have valves along their length?

(1)

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- ☒ **A** P
- ☒ **B** P and R
- ☒ **C** Q
- ☒ **D** Q and R

(iii) Which blood vessel does the coronary artery branch from?

(1)

- ☒ **A** P
- ☒ **B** Q
- ☒ **C** R
- ☒ **D** S

(b) Capillaries enable the movement of molecules between the blood and cells.

The distance the molecules diffuse between the blood and cells affects the rate of diffusion. This is the diffusion distance.

(i) Name **two** other factors that affect the rate of diffusion.

(1)

1 .....

2 .....

(ii) Name the law of diffusion that can be used to calculate the rate of diffusion.

(1)

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(c) The table shows the time taken for a molecule to diffuse over different diffusion distances.

Diffusion distance	Diffusion distance / μm	Time taken to diffuse the distance	Time taken to diffuse the distance / milliseconds
1 μm	1	0.5 milliseconds	0.5
10 μm	10	50.0 milliseconds	50.0
100 μm	100	5.0 seconds	
1 mm		8.3 minutes	

(i) Complete the table to show the missing values.

(2)

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(ii) Explain why it is important for body cells to be close to capillaries to meet the demands of respiration.
Use the information in the question and the table to support your answer.

(3)

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(Total for question = 10 marks)

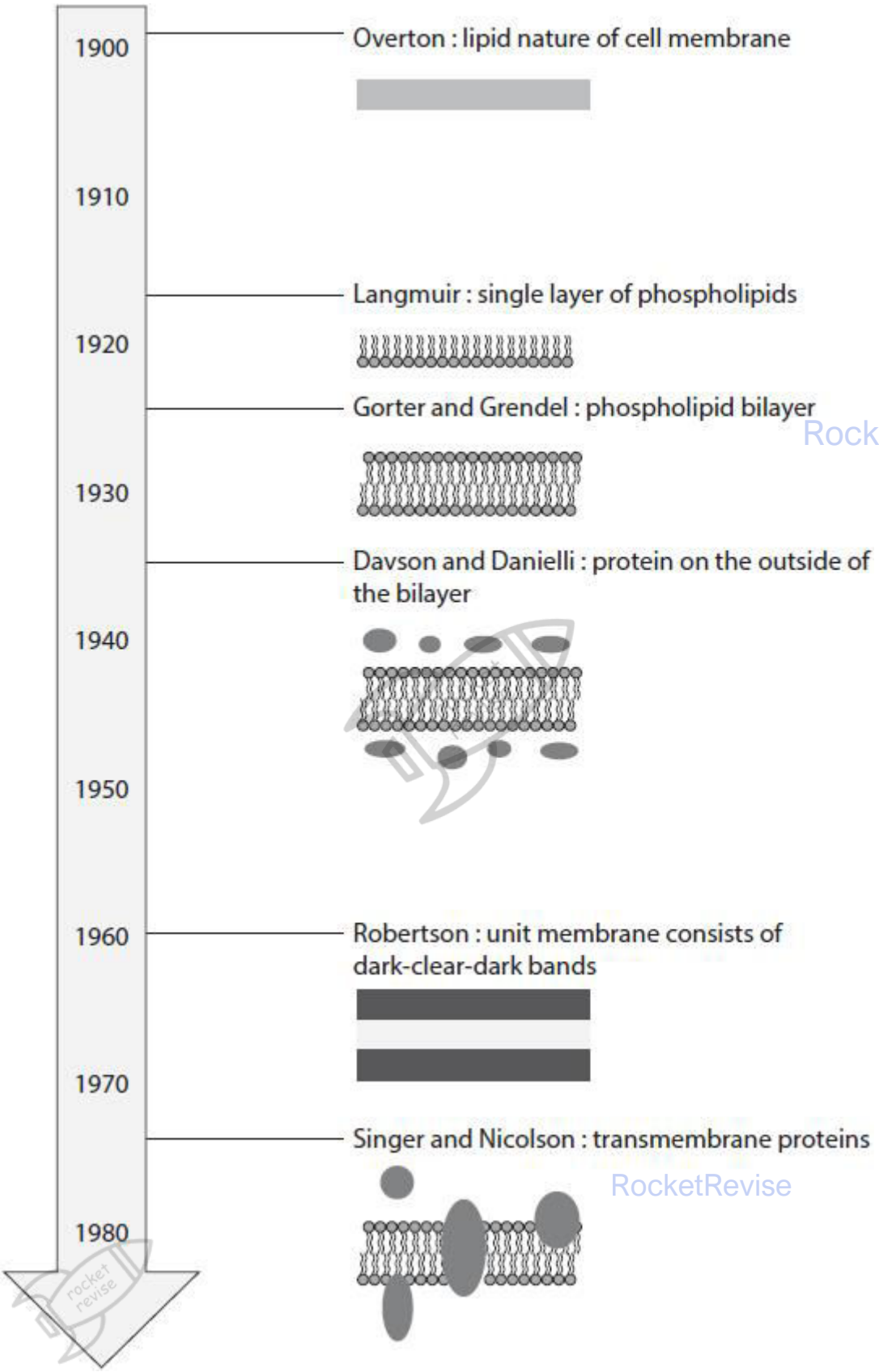
Q25.

The structure of the cell membrane has been modelled to interpret the data available at the time.

The diagram shows a timeline of six models proposed for the structure of the cell membrane.

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(a) Overton suggested that some types of solute could cross the lipid membrane, whereas other types of solute could not.

Explain the difference between these two types of solute and why only one type can cross the lipid membrane.

(2)

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(b) Langmuir's model was a single layer of phospholipids.

Explain why the phospholipids in a cell membrane cannot be arranged in a single layer.

(2)

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(c) Gorter and Grendel suggested that there were various types of phospholipid but they were all arranged in a bilayer.

Give **two** ways in which phospholipids can be different from each other.

(2)

1 .....

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2 .....

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(d) The Davson and Danielli model was the first to include proteins.

Explain why some membrane proteins would **not** be able to carry out their transport functions if they were located only on the surface of the phospholipid bilayer.

(2)

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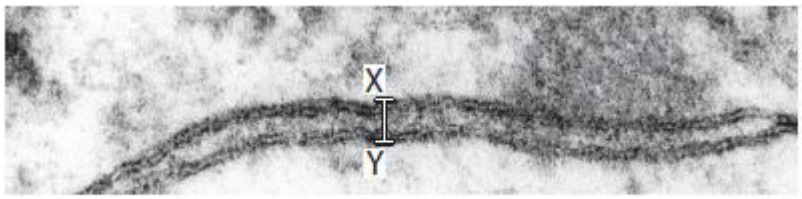
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(e) Robertson implied that there was no space between the phospholipid layers and that the thickness of the membrane could be measured.

The photograph shows part of two adjacent cell membranes.

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(Source: © DENNIS KUNKEL MICROSCOPY/SCIENCE PHOTO LIBRARY)

The thickness of these cell membranes between X and Y is 22.8 nm.
Calculate the magnification of this photograph.
Express your answer to **two** significant figures.

(2)

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Answer .....

(f) In 1972, Singer and Nicolson proposed the fluid mosaic model of the cell membrane.

Explain why their model has been given this name.

(2)

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(Total for question = 12 marks)

Q26.

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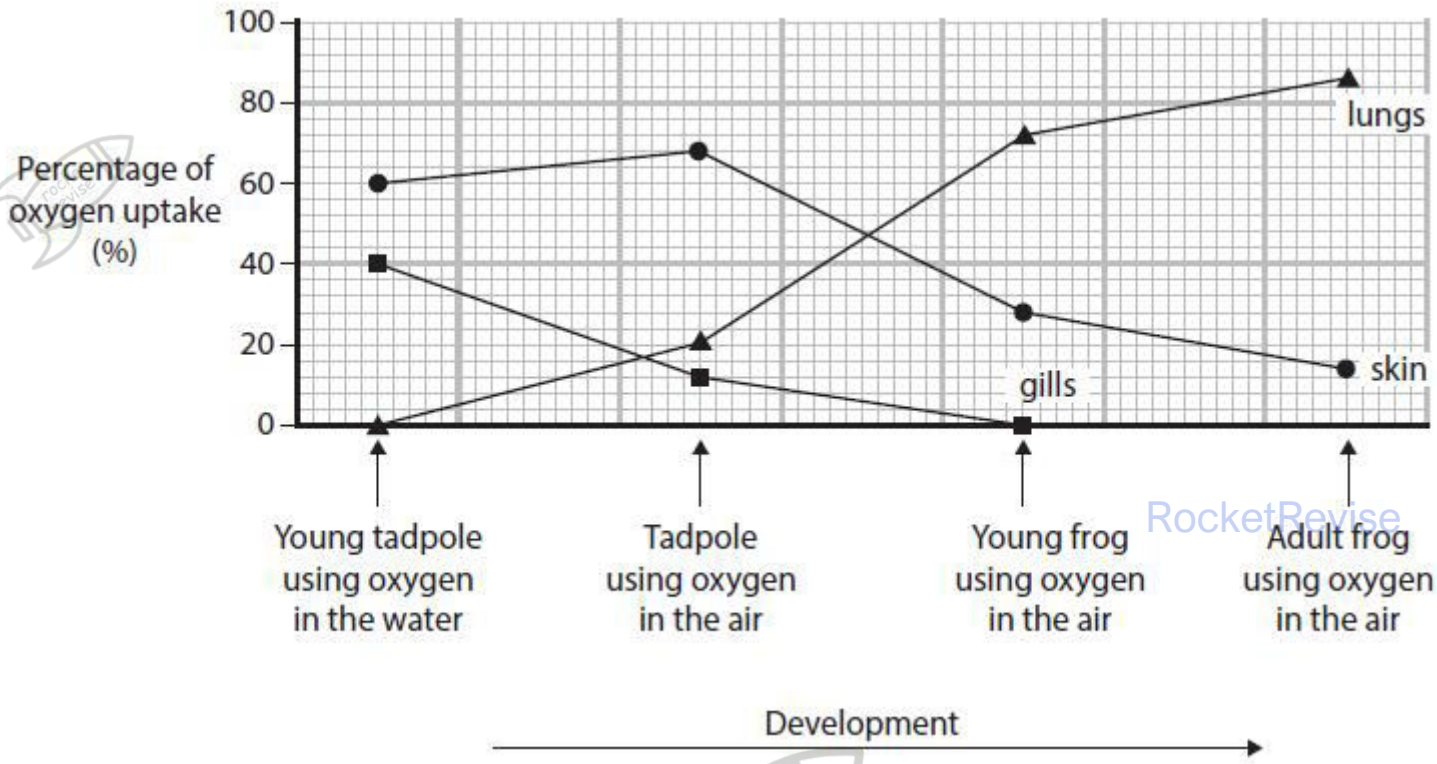
One function of the heart is to transport oxygen from the gas exchange surface to the cells.

(a) The gas exchange surfaces of a frog change as a tadpole develops into an adult frog.

The graph shows some of these changes.

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The contribution to gas exchange of the lungs, gills and skin changes as a tadpole develops into a frog.
Explain the importance of these changes.

(3)

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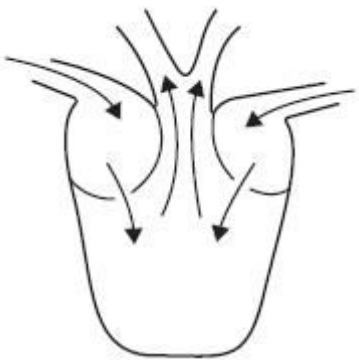
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(b) The diagram shows the structure of the heart of a frog.



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Give **one** similarity and **one** difference between the structure of this heart and a mammalian heart.

(2)

Similarity

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Difference

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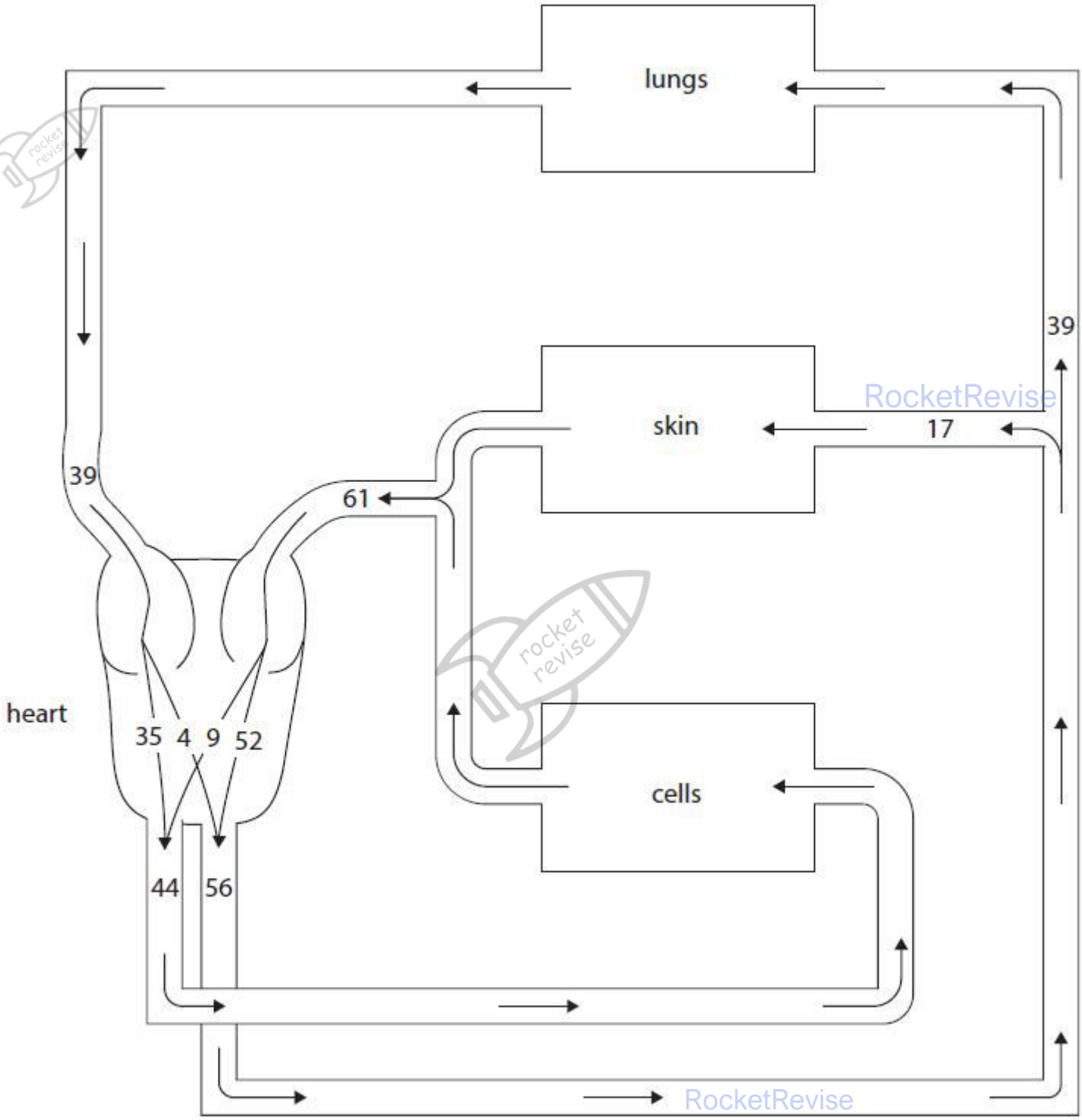
*(c) The diagram shows the distribution of blood flowing through the circulatory system of a frog.

The percentages of blood flowing through the parts of the circulatory system are shown on the diagram.

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The cells of the frog need oxygen to survive.
Explain how the circulatory system and the gas exchange surfaces of an adult frog are able to meet this need for oxygen.
Use the information in the diagram to support your answer.

(6)

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(Total for question = 11 marks)

Q27.

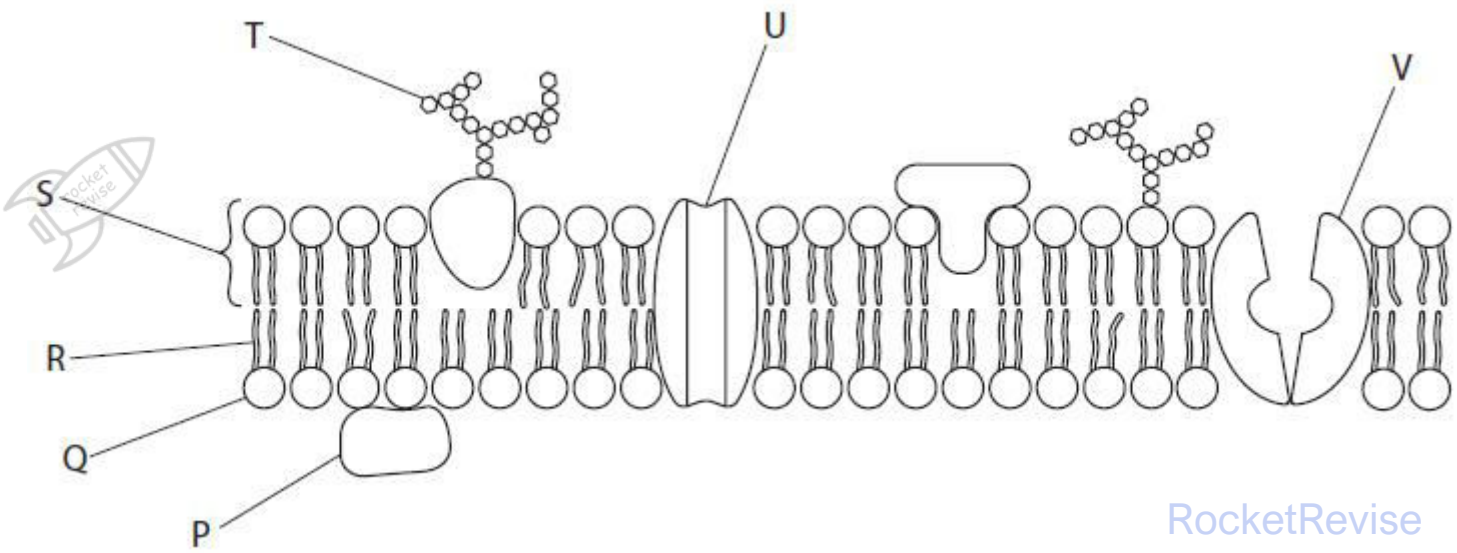
Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The structure and properties of a cell membrane determine which molecules can pass through the membrane.

(a) The diagram shows some structures in part of a cell membrane.

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(i) Which structure **prevents** large polar (hydrophilic) molecules from diffusing through the membrane?

(1)

- ☐ **A** Q
- ☐ **B** R
- ☐ **C** T
- ☐ **D** V

(ii) Which structure **allows** ions to diffuse through the membrane?

(1)

- ☐ **A** P
- ☐ **B** S
- ☐ **C** T
- ☐ **D** U

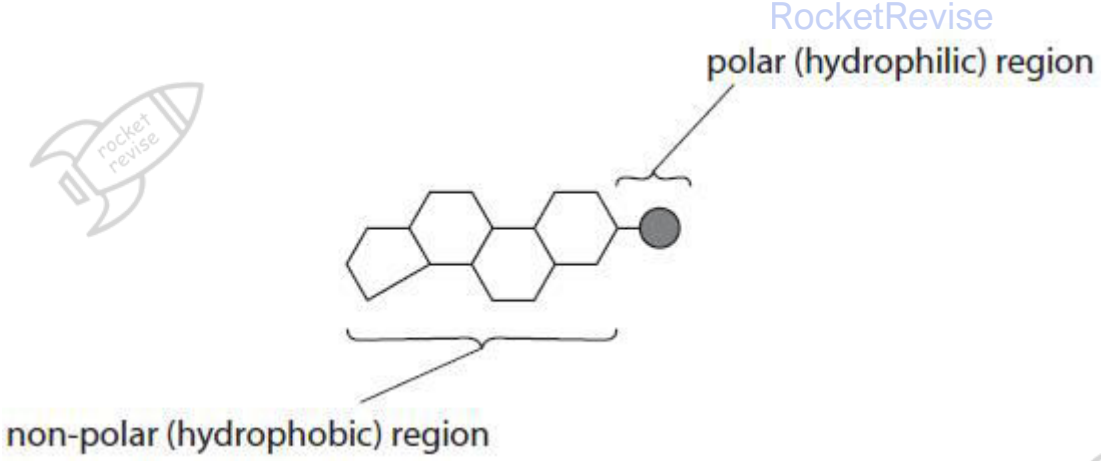
(iii) Which structure **allows** the membrane to be fluid?

(1)

- ☐ **A** P
- ☐ **B** Q
- ☐ **C** S
- ☐ **D** U

(b) The diagram shows a cholesterol molecule.

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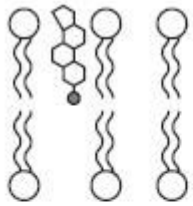


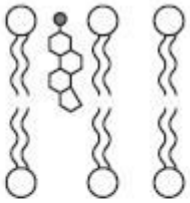
Which diagram shows the correct position of a cholesterol molecule in the membrane?

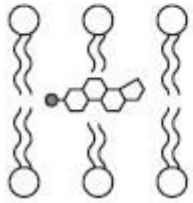
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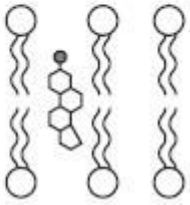
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☐ A 

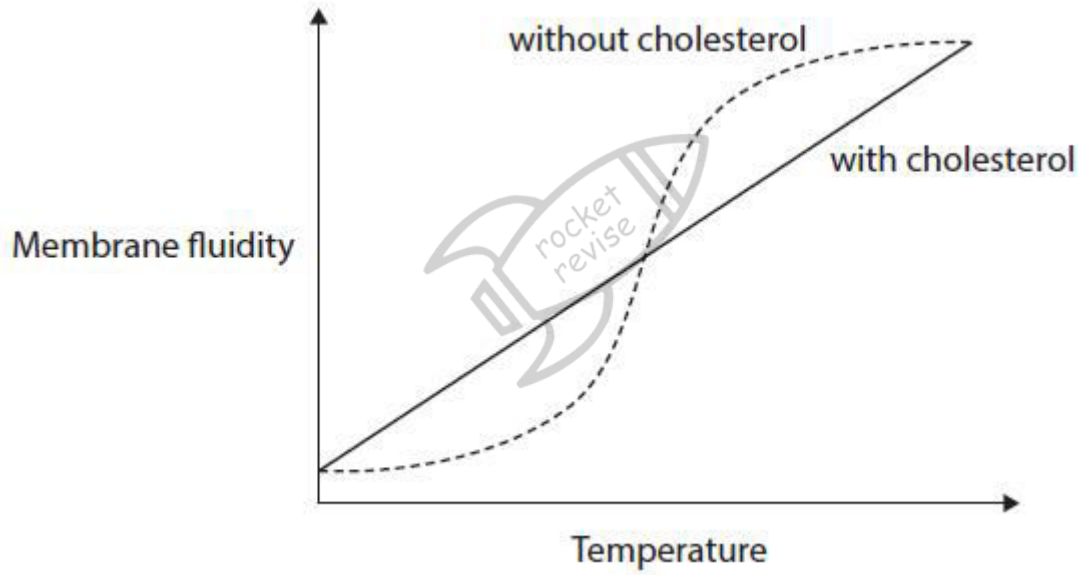
☐ B 

☐ C 

☐ D 

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(c) The graph shows the effect of temperature and cholesterol on membrane fluidity.



Describe the effect of temperature and cholesterol on membrane fluidity.

(3)

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(d) The table shows the composition of protein, lipid and carbohydrate in three membranes, A, B and C.

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Membrane	Percentage of protein (%)	Percentage of lipid (%)	Percentage of carbohydrate (%)
A	49	43	8
B	54	42	4
C	75	25	0

(i) Calculate the ratio of protein : lipid in membrane A.
Give your answer to **2** decimal places.

(1)

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Answer ..... : 1

(ii) Compare and contrast the composition of these three membranes.

(3)

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(Total for question = 11 marks)

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